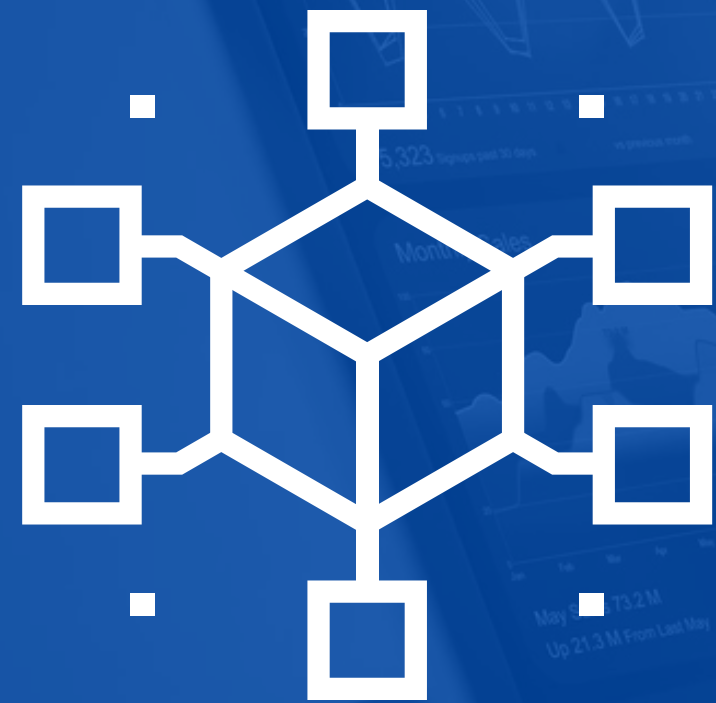




[Link Project](#)



Analyze Customer Personality to Enhance Marketing Campaigns Using Machine Learning Clustering

By: Aldi Vibriani

Overview

A company can experience rapid growth by understanding customer personalities, allowing it to provide better services and benefits to customers who have the potential to become loyal. By analyzing historical marketing campaign data to improve performance and target the right customers to transact on the company's platform, our focus is to develop a predictive clustering model. This model will assist the company in making informed decisions.



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Exploratory Data Analysis

Explore Insights

Discover patterns and trends to drive strategic decisions. Our exploratory data analysis reveals key insights for targeted marketing and customer engagement.

About Dataset:

This dataset consist of customer's information, number of products brought in last 2 years, customer's respond on campaigns, and where they purchase the products.

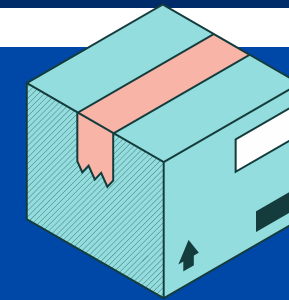
Customer's Information

- ID
- Year_Birth
- Education
- Marital_Status
- Income
- Kidhome
- Teenhome
- DT_Customers
- Recency
- Complain
- Z_CostContact
- Z_Revenue



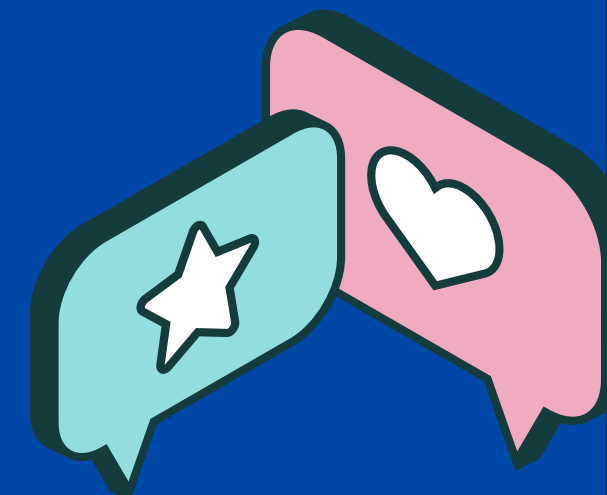
Products

- MntFruits
- MntMeatProducts
- MntFishProducts
- MntSweetProducts
- MntGoldProds



Promotion

- NumDealsPurchases
- AcceptedCmp1
- AcceptedCmp2
- AcceptedCmp3
- AcceptedCmp4
- AcceptedCmp5
- Response



Place

- NumwebPurchases
- NumCatalogPurchases
- NumeStorePurchases
- NumWebVisitsMonth



The majority of customers are middle-age adults

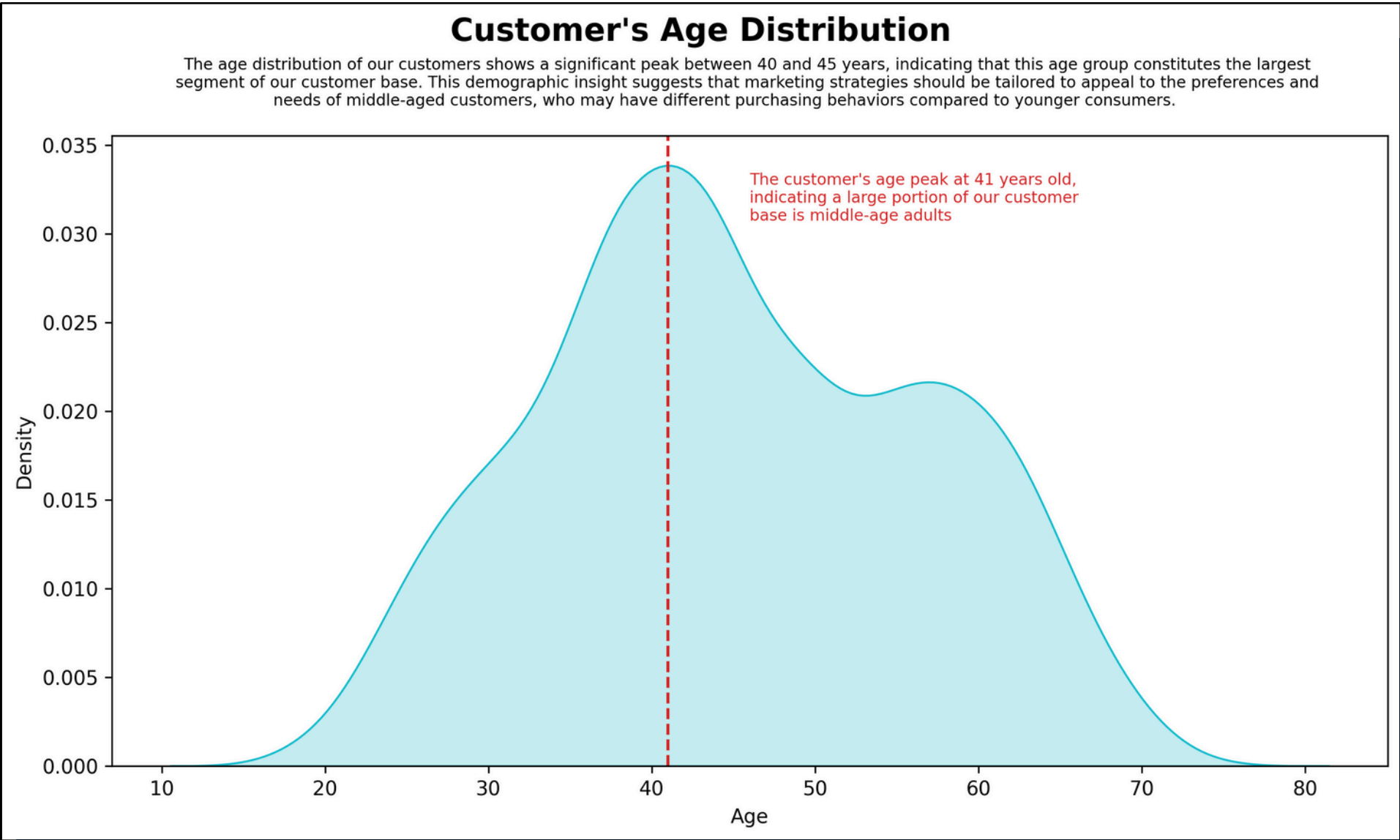


Image 1.1 - Customer's Age Distribution

Most of customers
have a Bachelor's
Degree

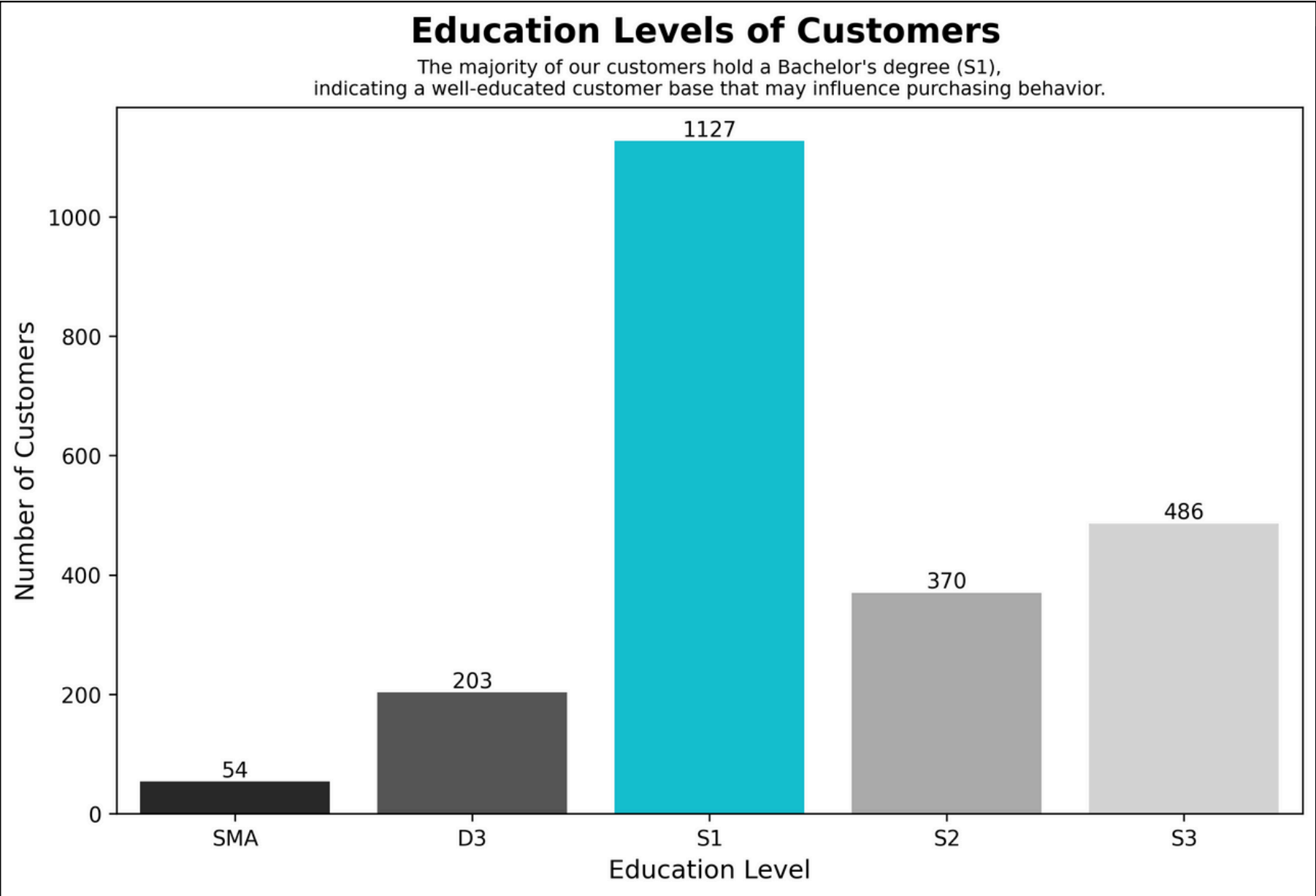


Image 1.2 - Education Levels of Customers

The majority of our customers are married

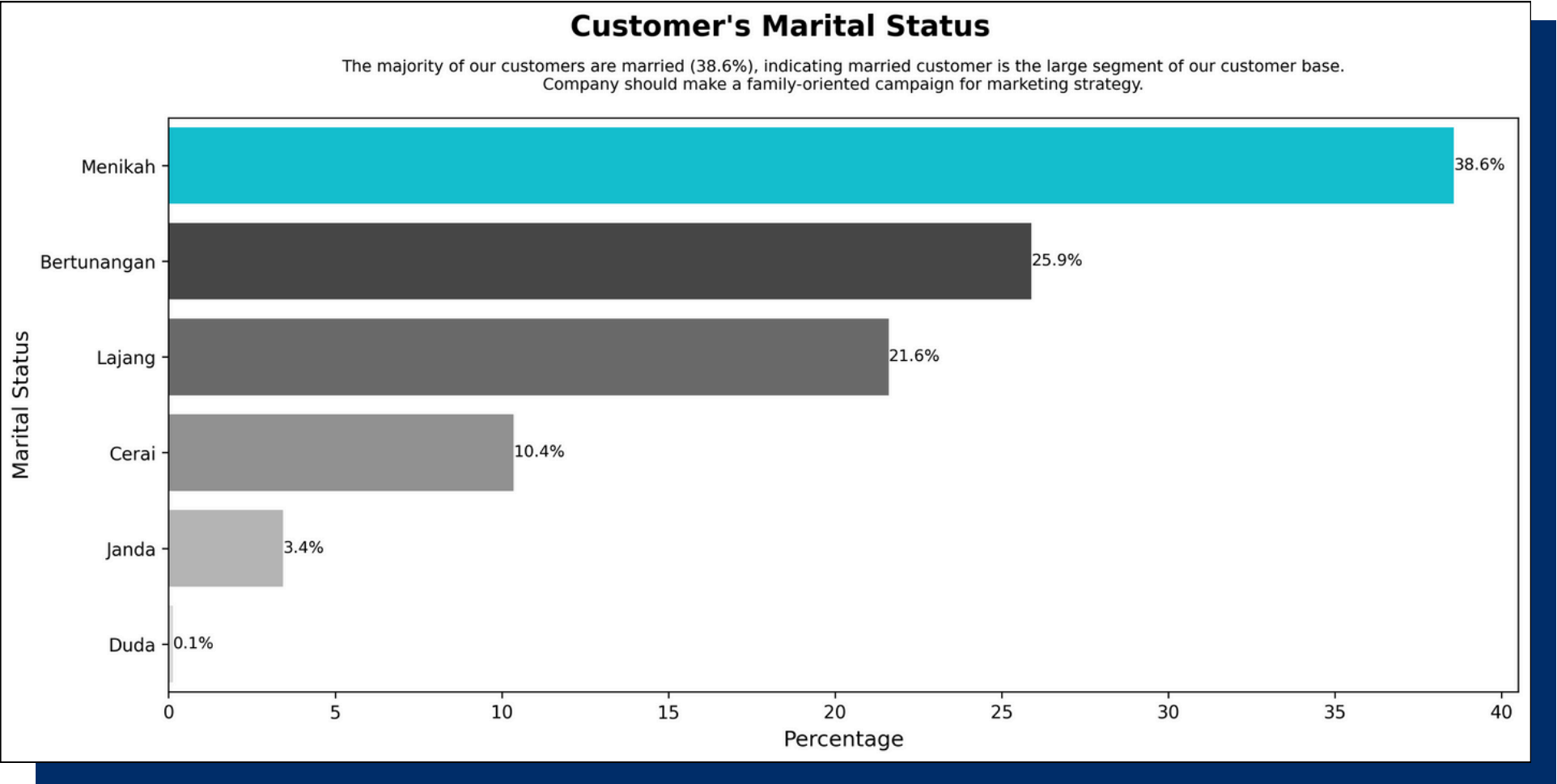


Image 1.3 - Customer's Marital Status

Our customers have at least one child

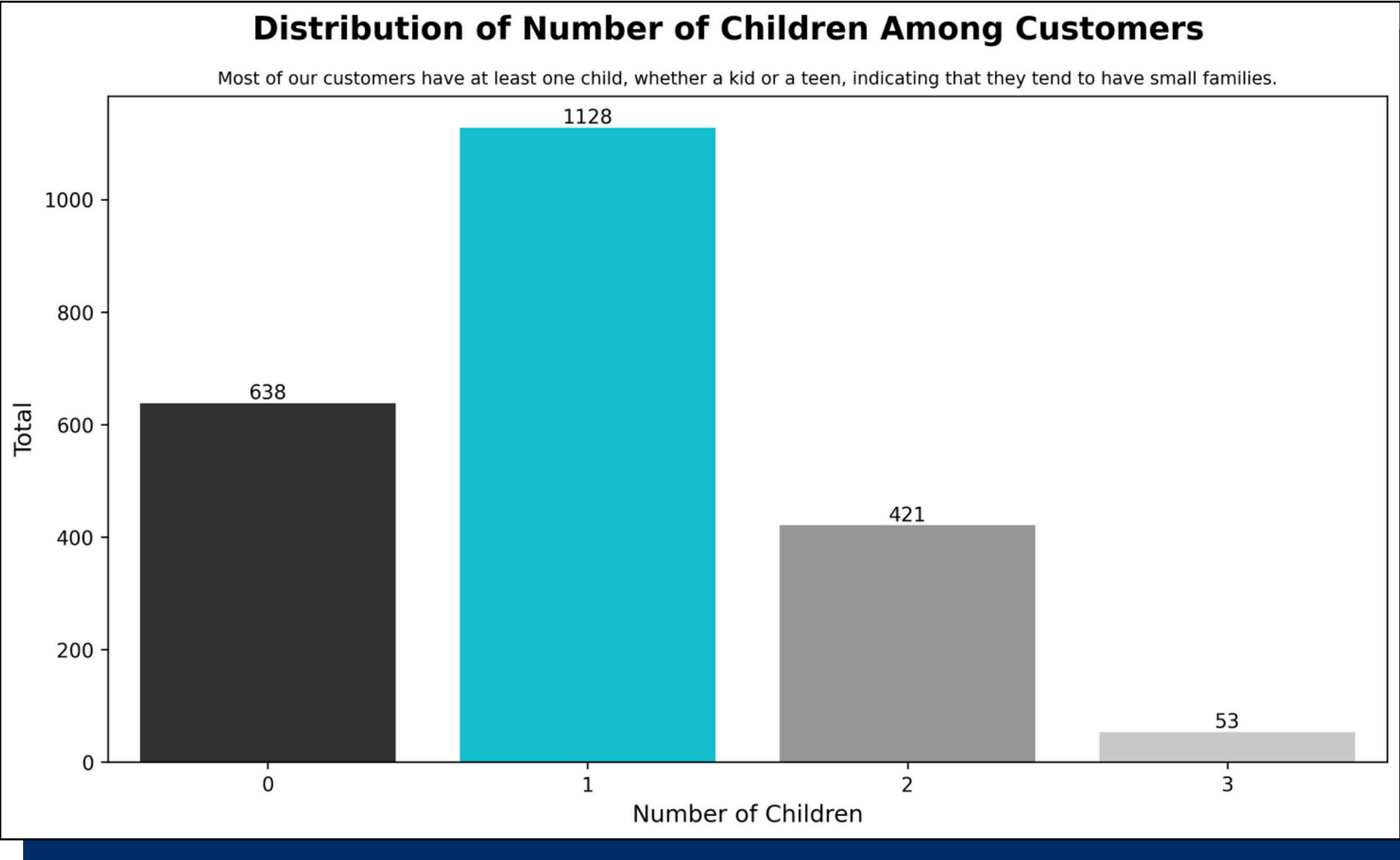


Image 1.4 - Number of Children Among Customers

There is a a notable outlier with an income of 666 million.

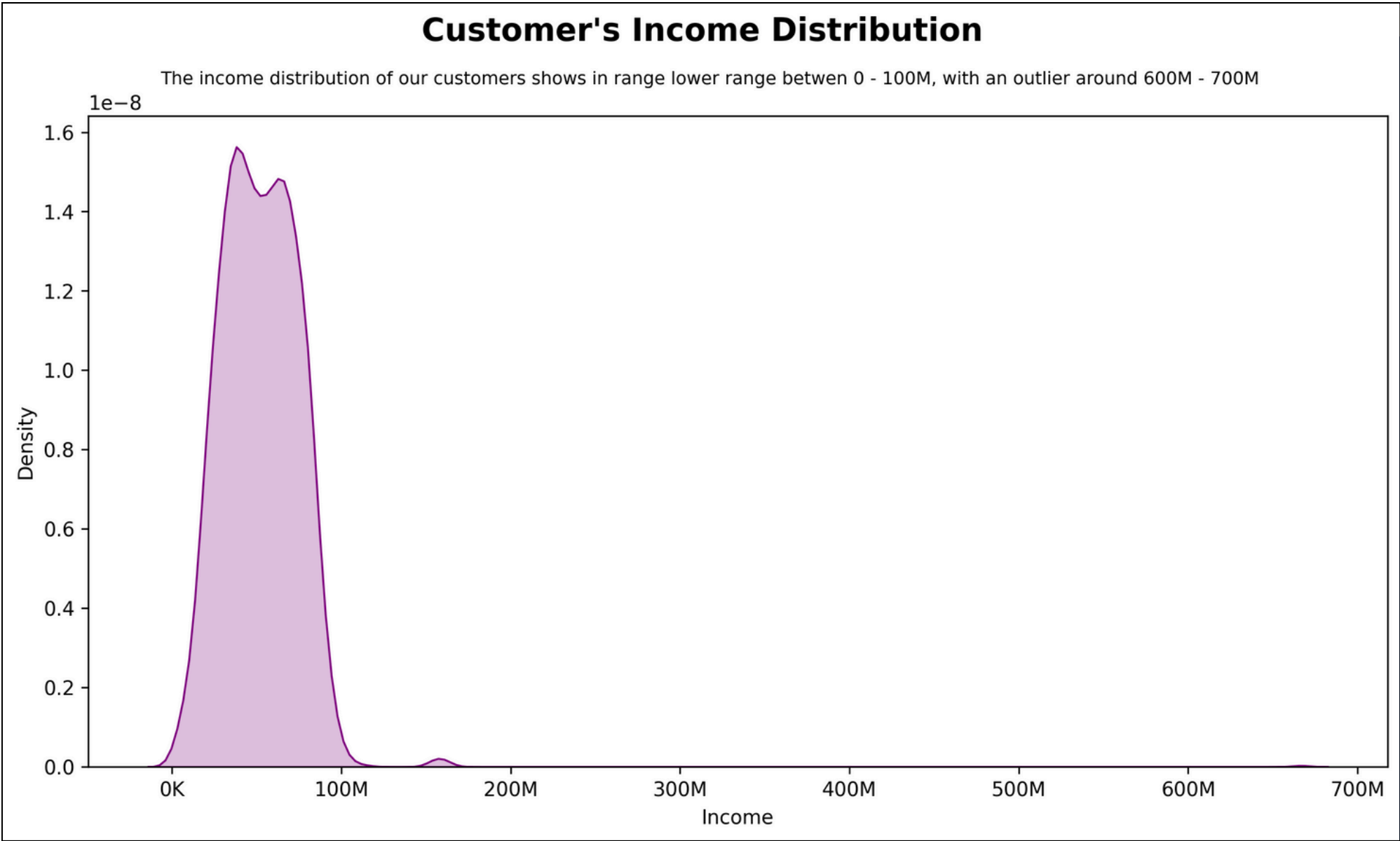


Image 1.5 - Customer's Income Distribution

**There is a presence
of two distinct
groups, indicating
potential
segmentation by
economic class**

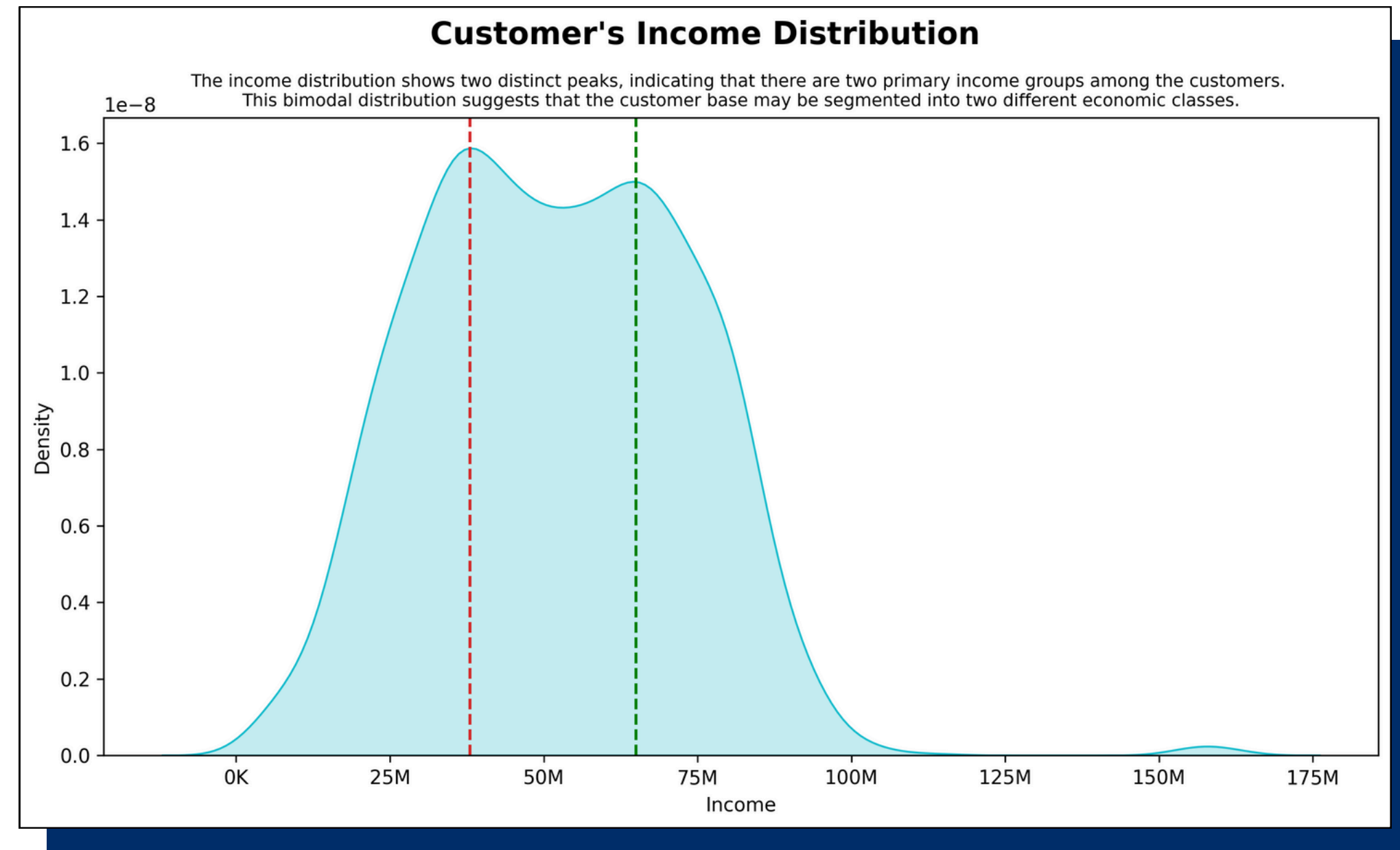


Image 1.6 - Customer's Income Distribution (Excluding outlier)

Approximately 31%
of our customers
have made a
purchase in the last
30 days

	Class	f	mp	rf	cf	percentage
0	0-4	136	2.0	0.060714	136	0.06
1	5-9	97	7.0	0.043304	233	0.10
2	10-14	122	12.0	0.054464	355	0.16
3	15-19	101	17.0	0.045089	456	0.20
4	20-24	111	22.0	0.049554	567	0.25
5	25-29	125	27.0	0.055804	692	0.31
6	30-34	111	32.0	0.049554	803	0.36
7	35-39	103	37.0	0.045982	906	0.40
8	40-44	91	42.0	0.040625	997	0.45
9	45-49	125	47.0	0.055804	1122	0.50
10	50-54	124	52.0	0.055357	1246	0.56
11	55-59	110	57.0	0.049107	1356	0.61
12	60-64	108	62.0	0.048214	1464	0.65
13	65-69	105	67.0	0.046875	1569	0.70
14	70-74	120	72.0	0.053571	1689	0.75
15	75-79	104	77.0	0.046429	1793	0.80
16	80-84	122	82.0	0.054464	1915	0.85
17	85-89	107	87.0	0.047768	2022	0.90
18	90-94	115	92.0	0.051339	2137	0.95
19	95-99	103	97.0	0.045982	2240	1.00

Image 1.7 - Recency Class Distribution

We have a low loyal customers, about 70% of customers not made repeat order within 30 days

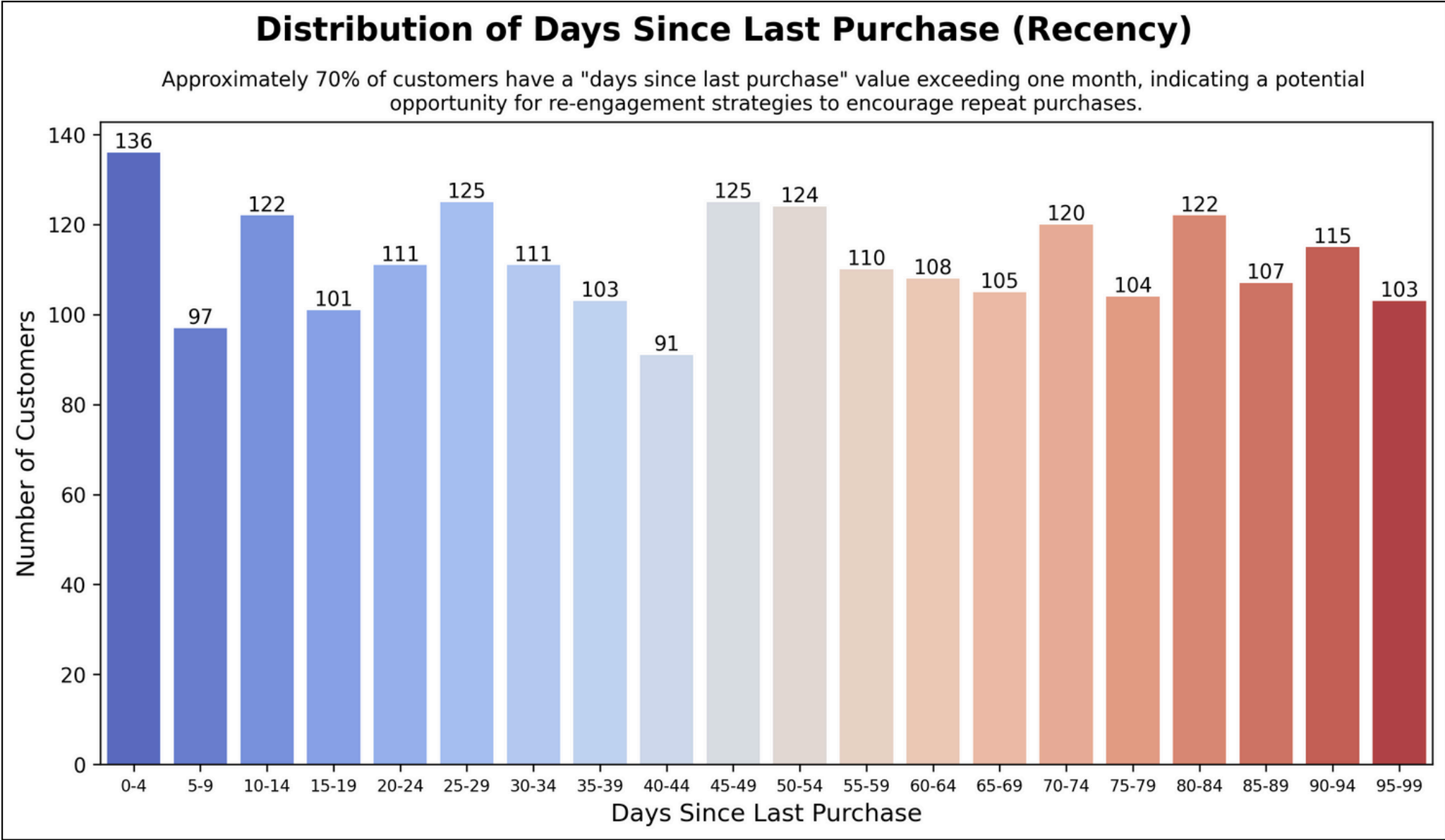


Image 1.8 - Distribution of Days Since Last Purchase

The majority of our customers do not accept the offer from any campaign.

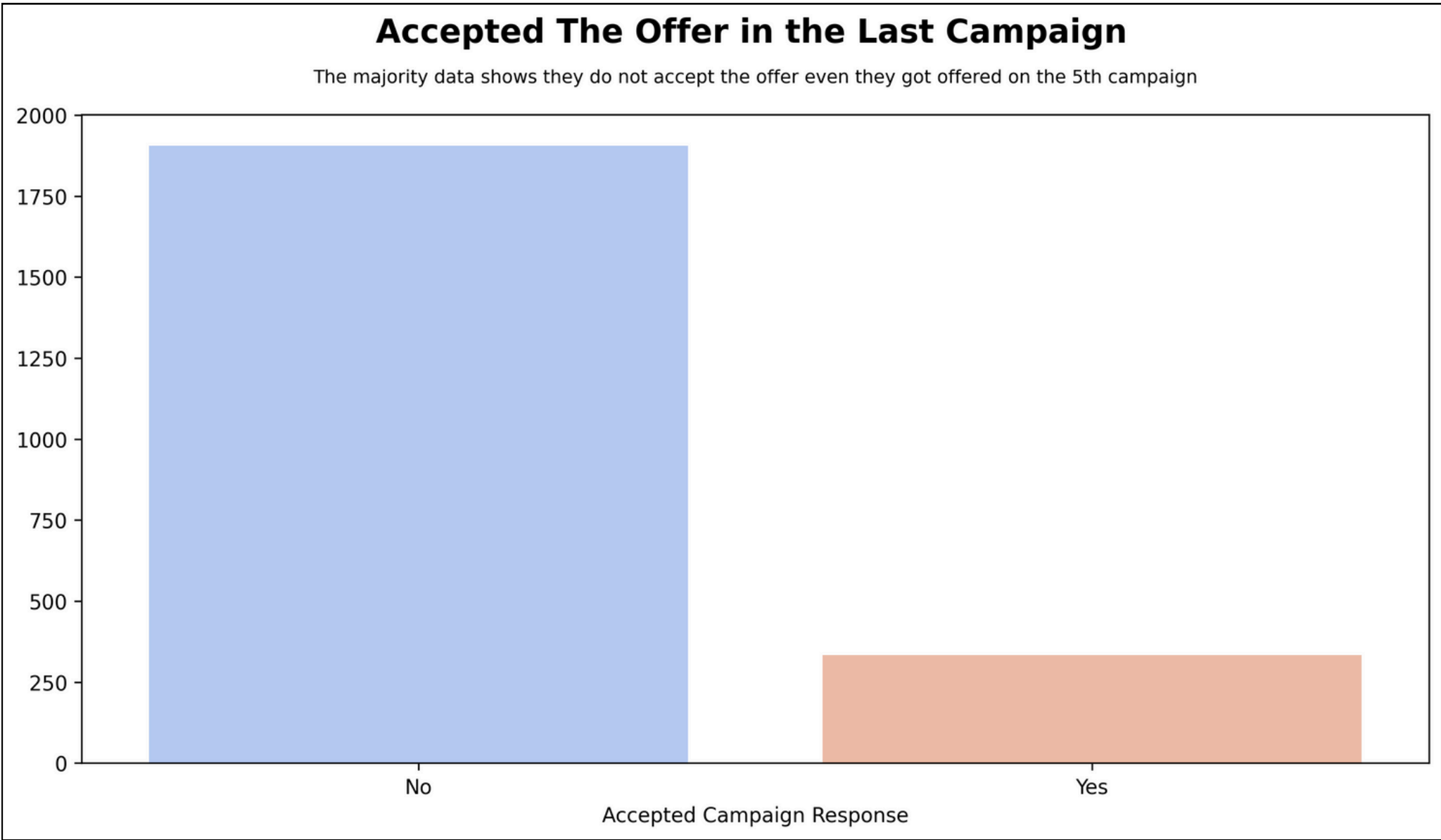


Image 1.9 - Accepted Offer from Campaign

None of our marketing campaigns are effective, especially the second campaign.

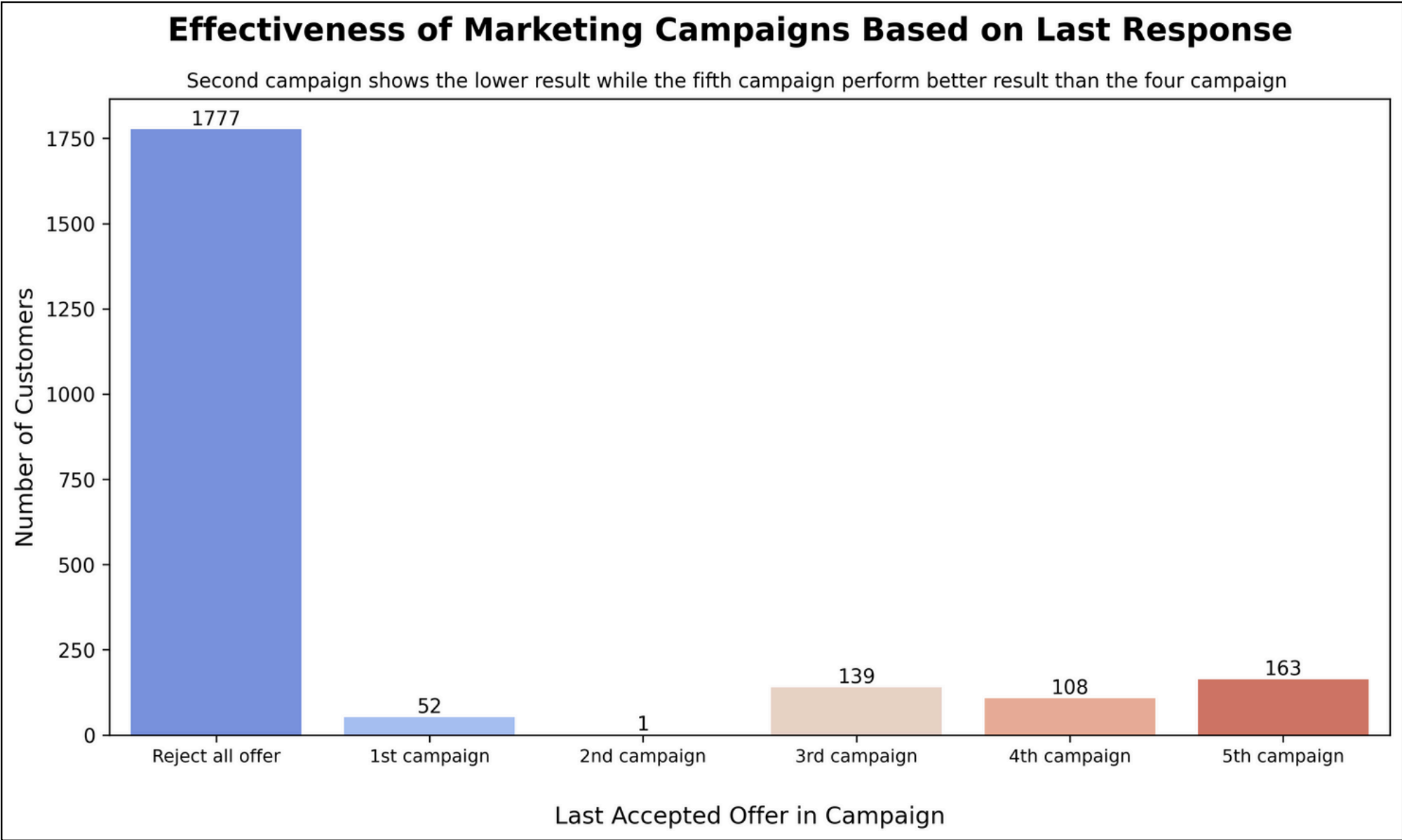


Image 1.10 - Marketing Campaign Effectiveness

There is low success
in driving repeat
purchases through
discount programs

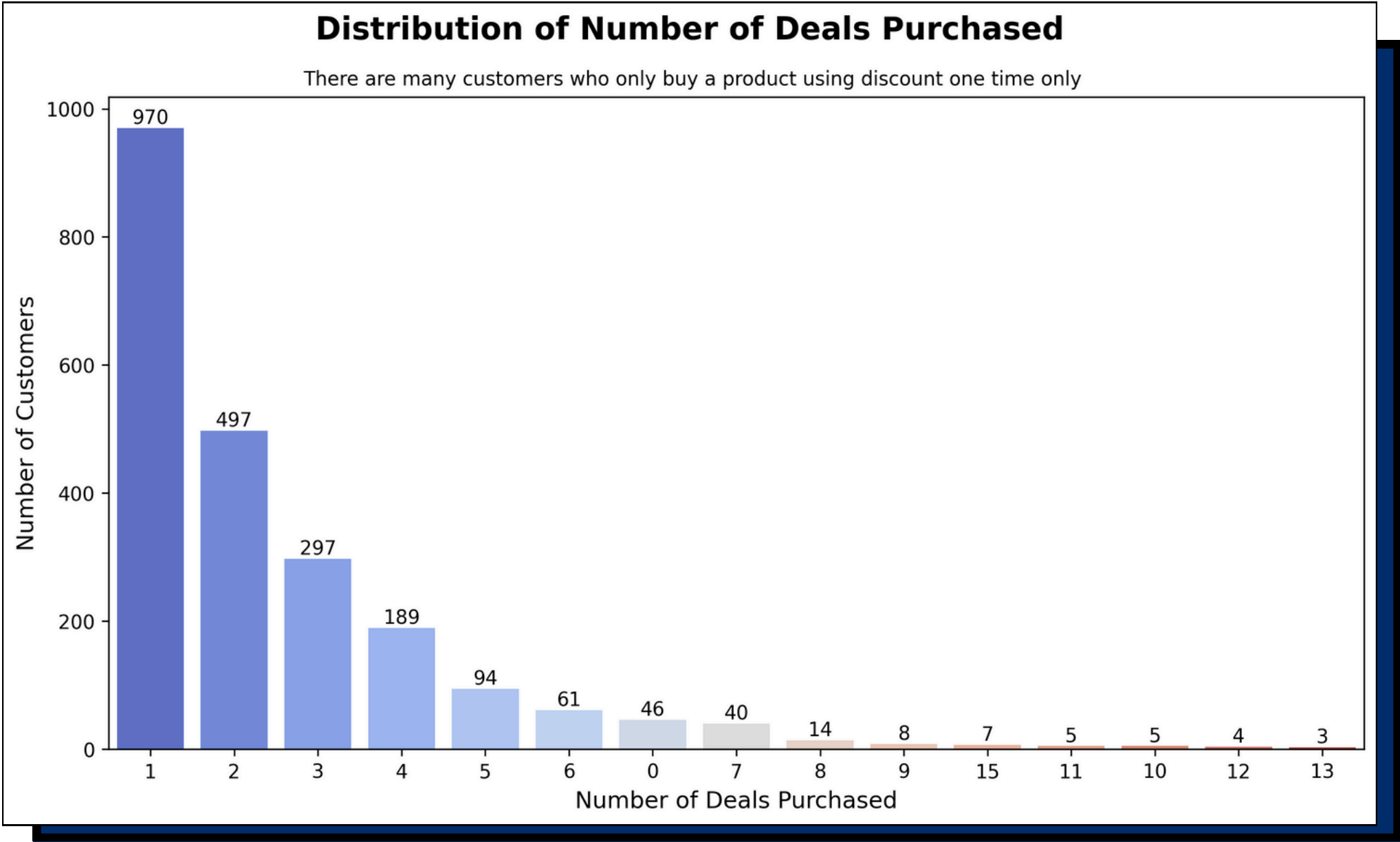


Image 1.11 - Number of Deals Purchased

Highest-income customers demonstrate surprisingly low spending across all product categories

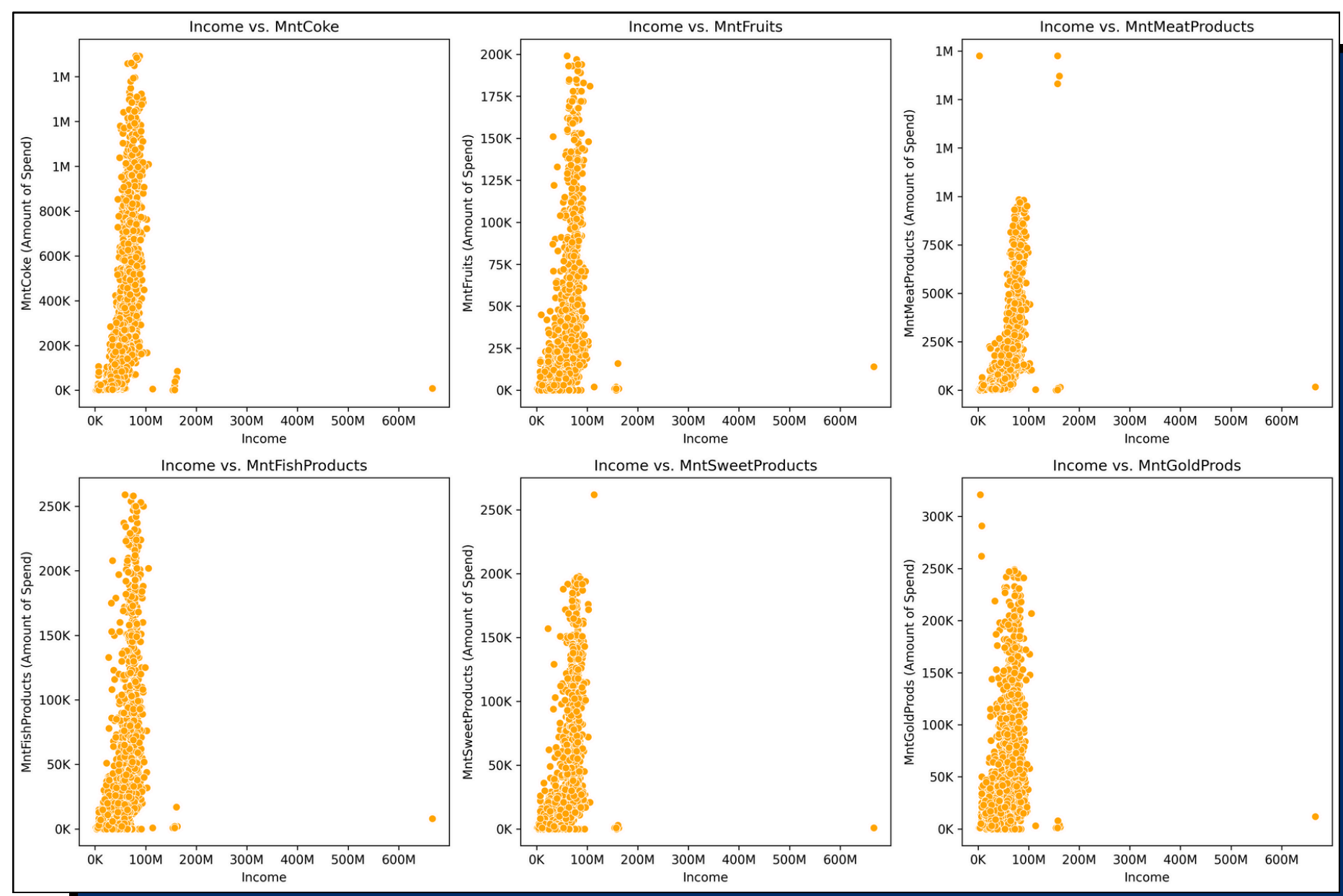


Image 1.12 - Purchases on Categories by Income

Purchase values
consistently rise in
proportion to
customer income
levels

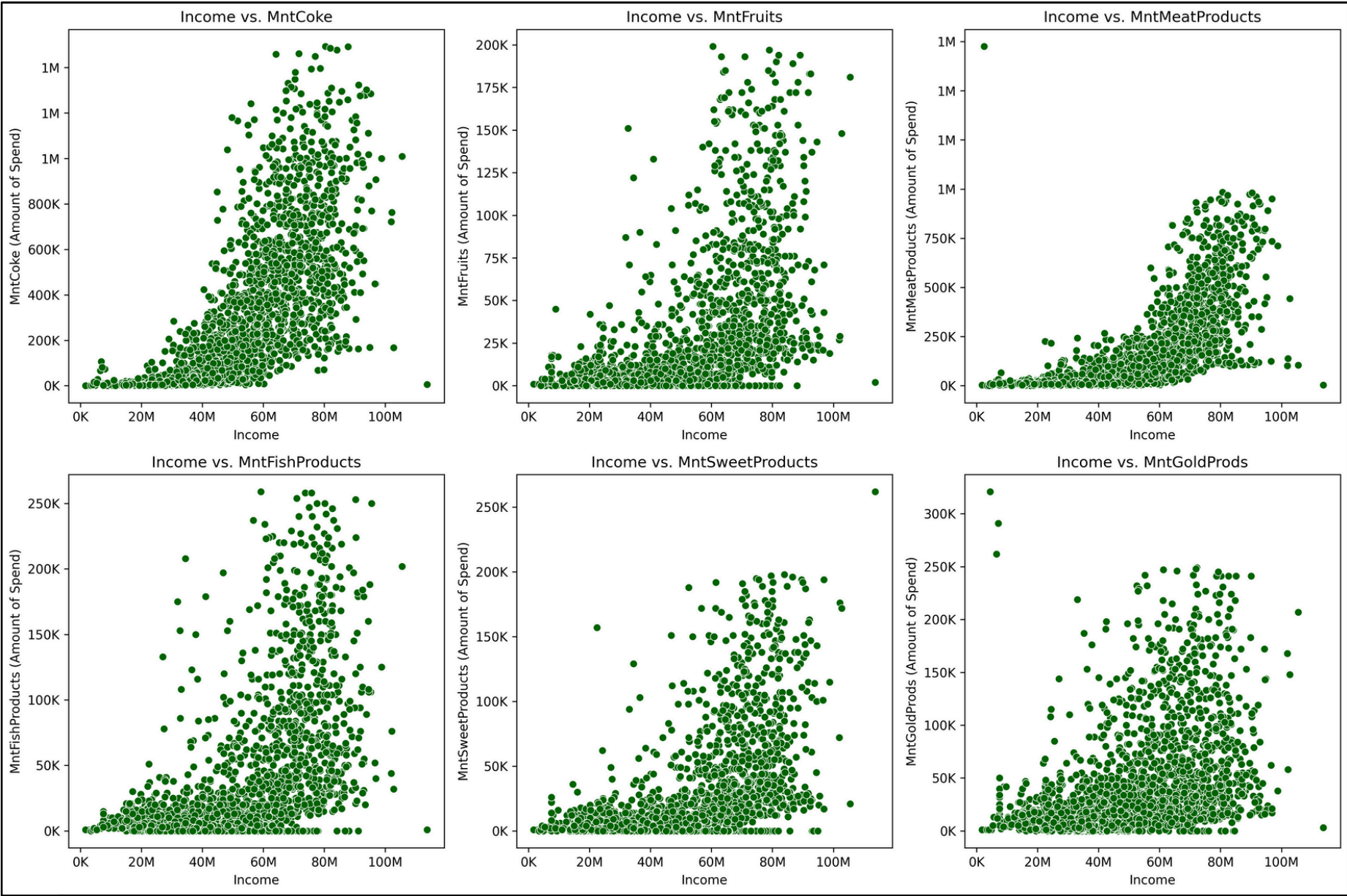


Image 1.13 - Purchases on Categories by Income (Drop Outliers)

Customers tend to purchase on offline store instead on website and catalogue

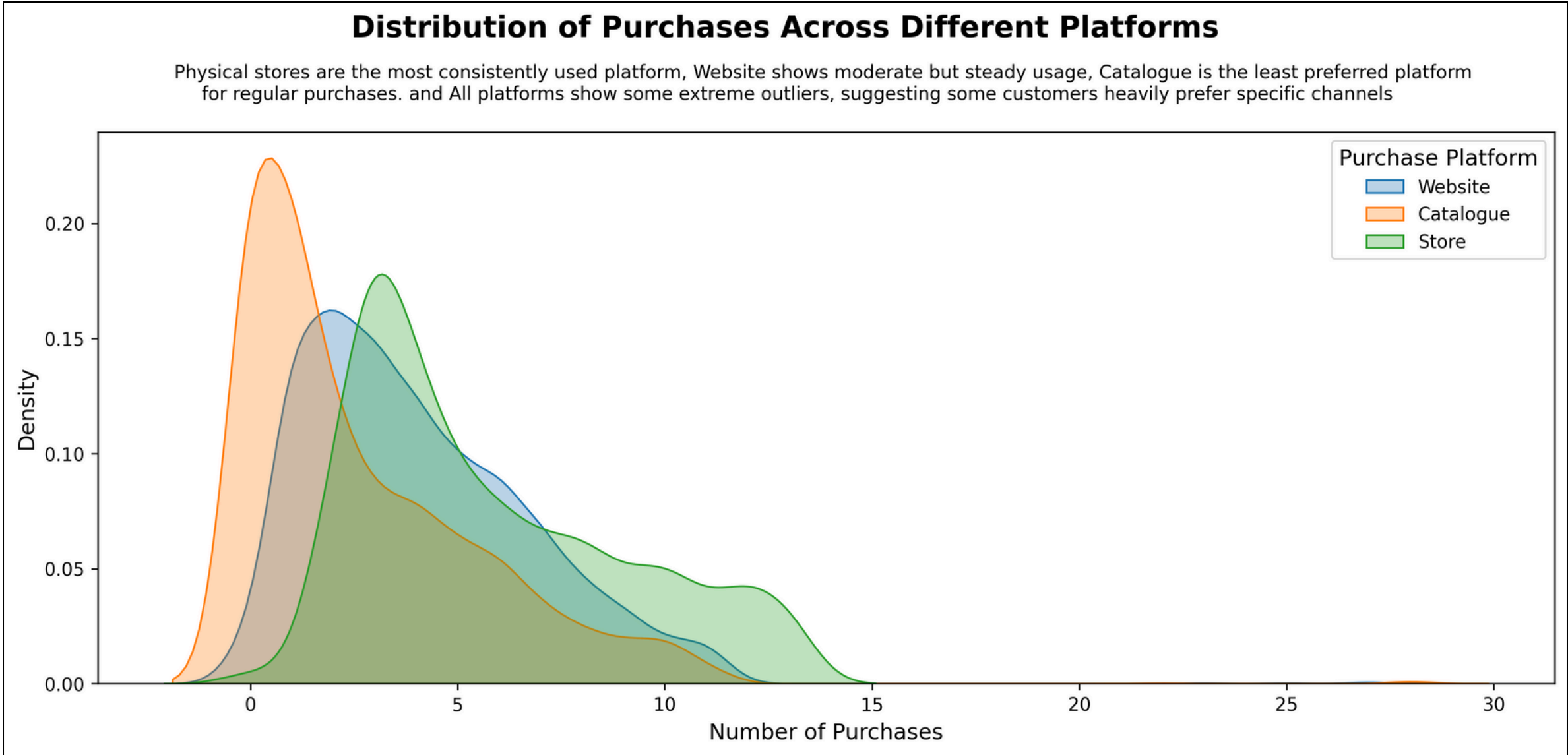


Image 1.14 - Purchase Platform

Income shows a positive correlation with purchase spending across all product categories while number of kids and teens show negative correlation

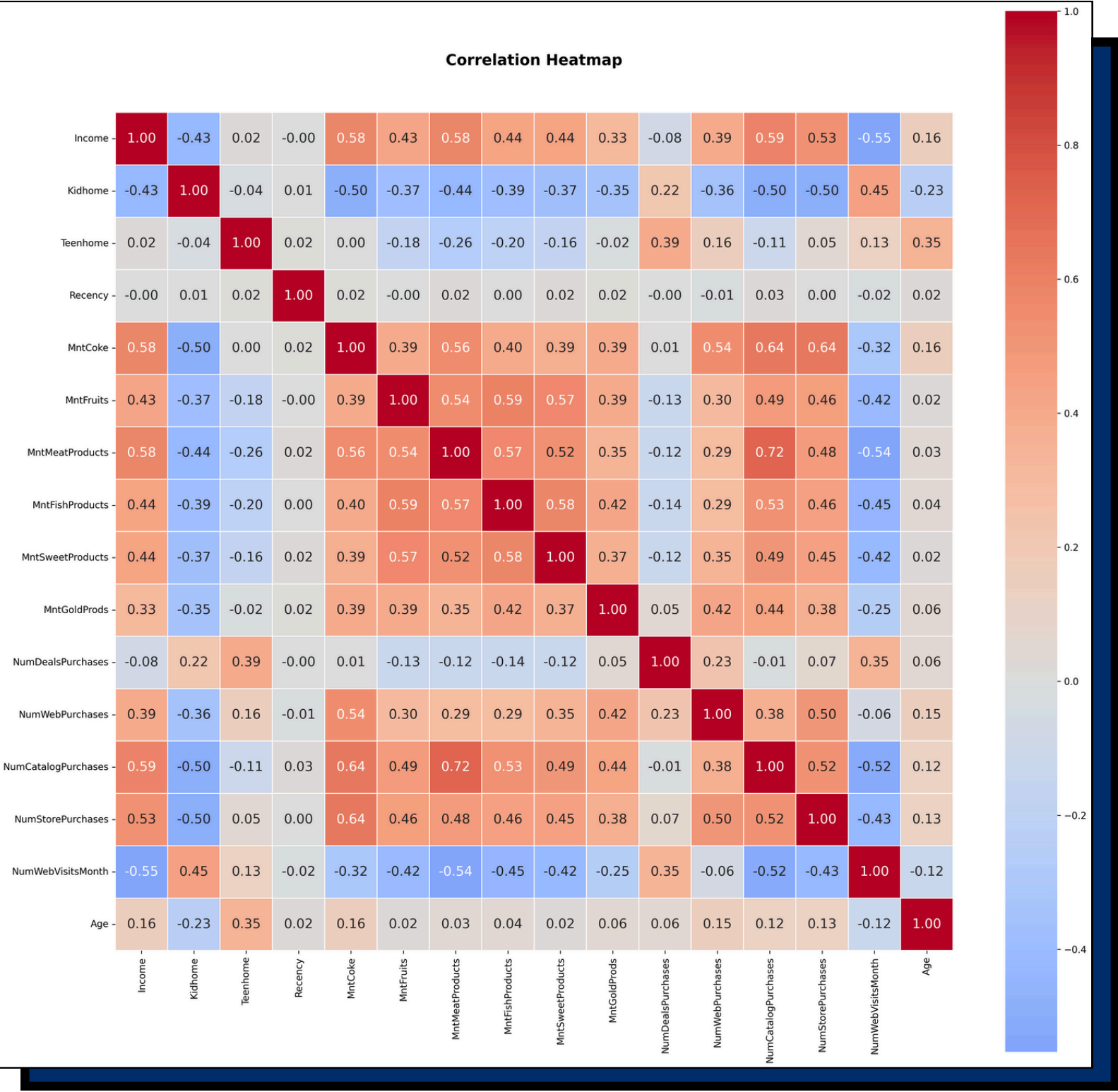


Image 1.15 - Correlation Heatmap

[Link project](#)

[Source Code](#)
[Preprocessing](#)

Data Preprocessing

Cleaning data for modelling

Handling missing value, duplicated data, outliers, feature encoding, and feature engineering

Drop Features

Code:

```
def check_unique_value(df):  
    for i in df.columns:  
        print(f"{i} \n {np.sort(df[i].unique())} \n")  
  
check_unique_value(df)
```

Output:

```
Complain  
[0 1]  
  
Z_CostContact  
[3]  
  
Z_Revenue  
[11]  
  
Response  
[0 1]
```

There are two features with one unique value, which is **Z_CostContact** and **Z_Revenue**.

These features will have no information for machine learning, so the suggestion is to drop these.

Handling Missing Values

"Income" Feature

24 or 1.07%
missing values

There are 24 or 1.07% missing values on "Income" feature.

We want to fill this missing values with **median** because the missing value is under 5%, meaning there will not change a whole data and median is robust from outliers.

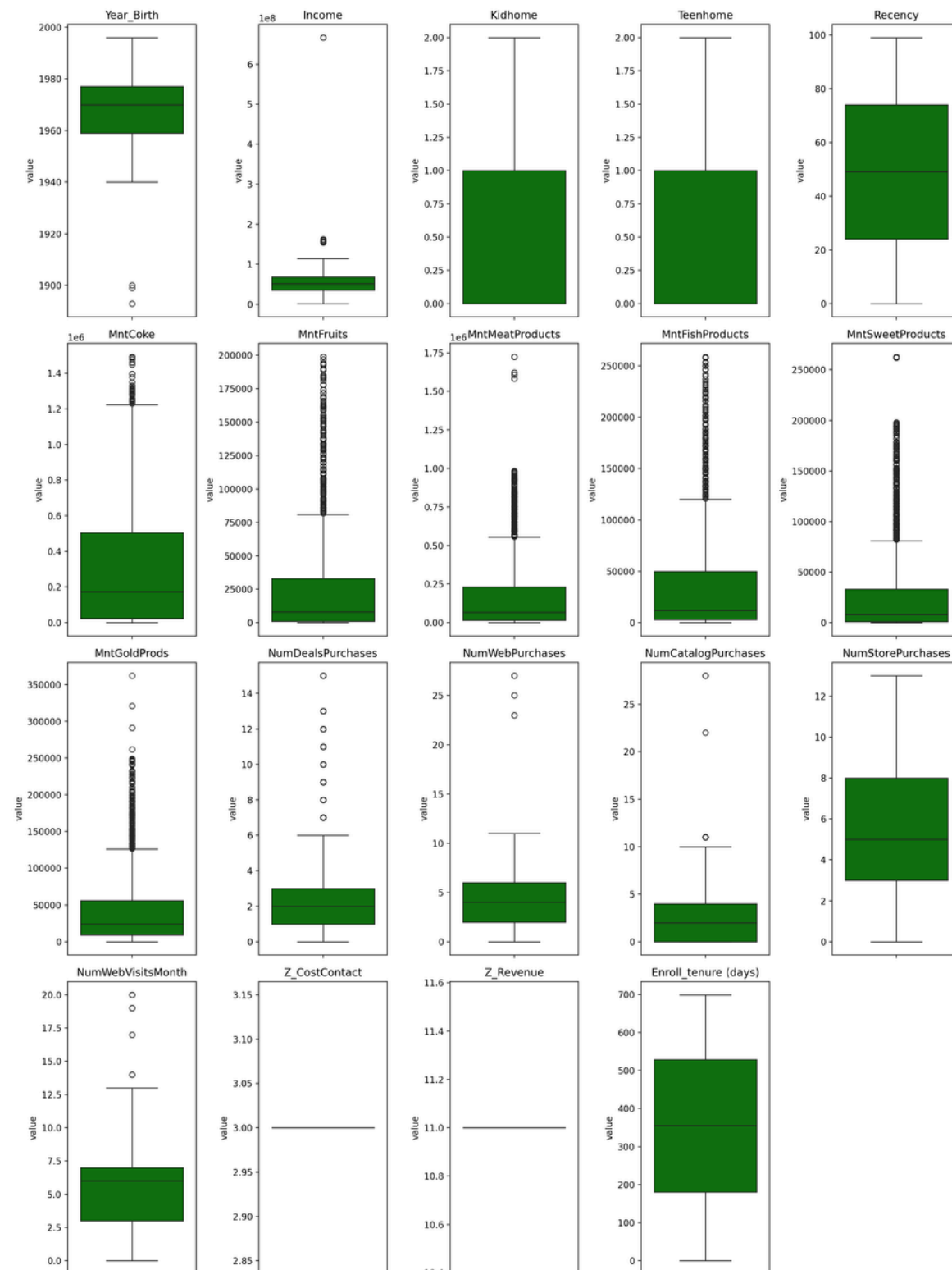
Handling Duplicated Data

Total Duplicates

183
duplicated data

We found 183 duplicated data in the dataset. So, we will **drop** them and keep the first data.

Handling Outliers



After checking the outliers, we found:

- There is an **extreme outlier** in “**Income**” feature.
- **Products purchase** (MntCoke, MntFruits, etc.) have outliers.
- Year_Birth, NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, and NumWebVisitsMonth shows outliers but the values looks like **not anomaly**, suggesting not to drop the outliers.

Handling Outliers - Income feature

Code:

```
df_transform.loc[df_transform['Income'].nlargest(10).index]
```

Output:

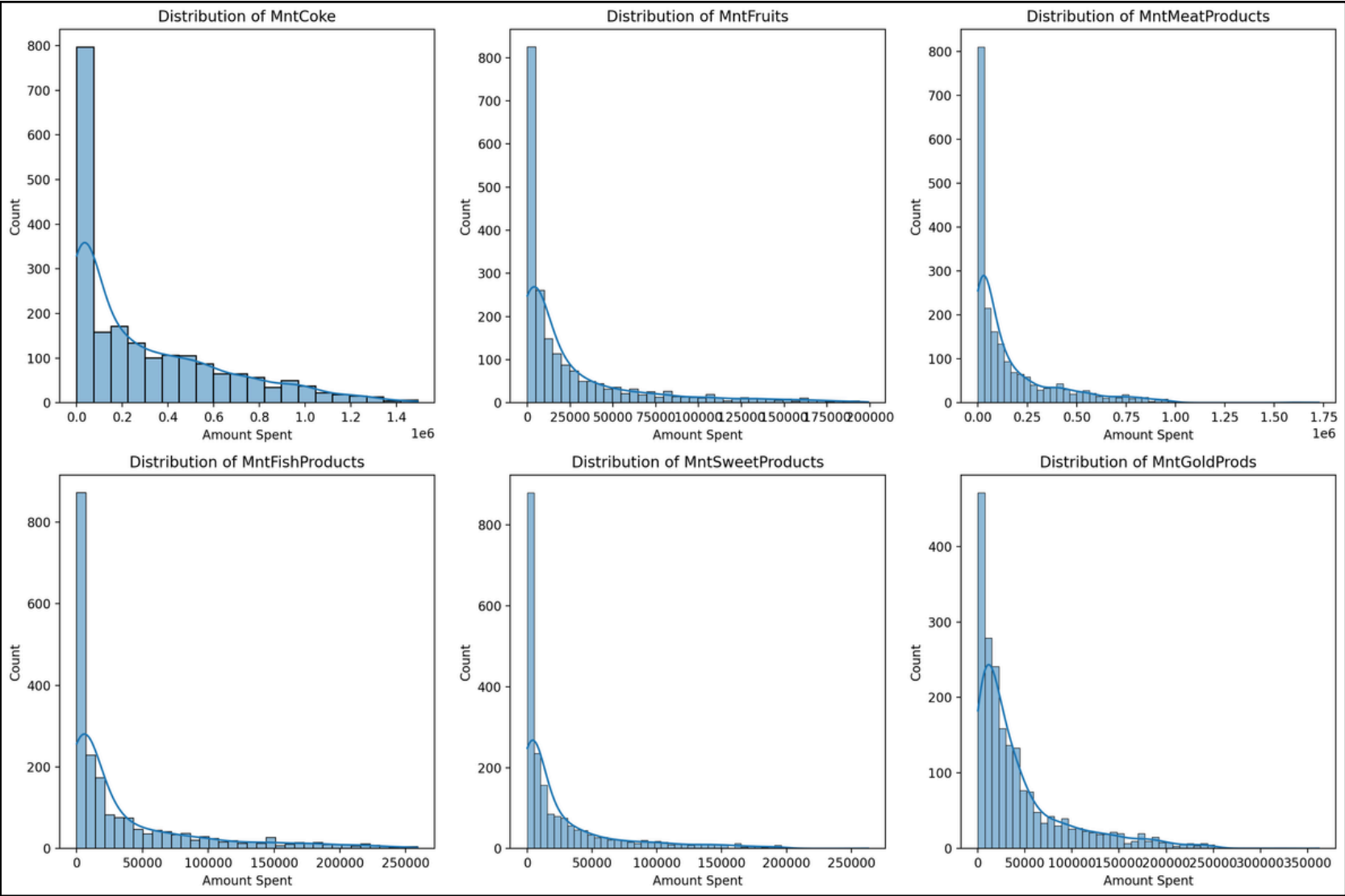
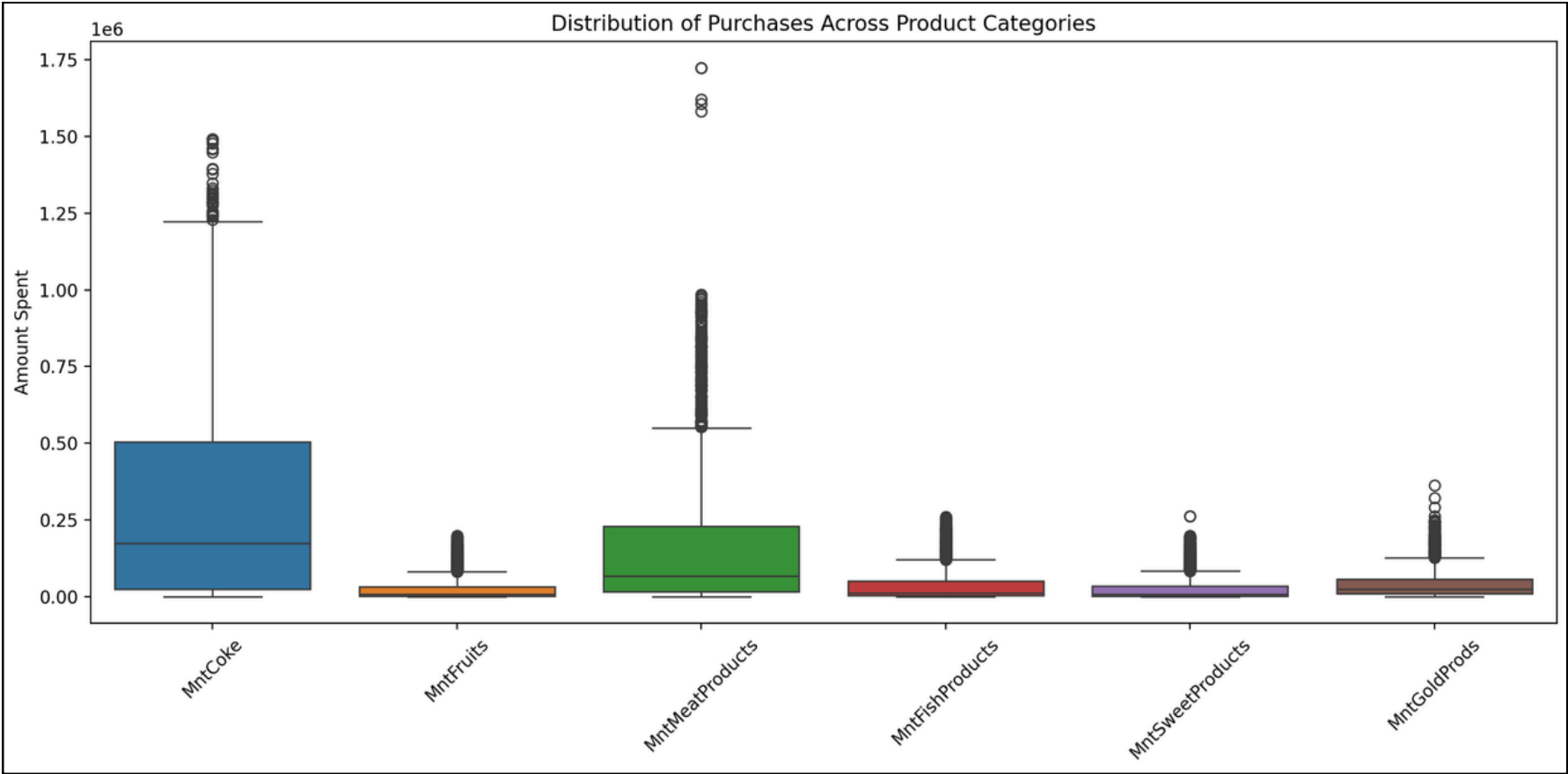
	Year_Birth	Education	Marital_Status	Income	Kidhome	Teenhome	Recency	MntCoke	MntFruits	MntMeatProducts	I
2233	1977	S1	Bertunangan	666666000.0	1	0	23	9000	14000	18000	
617	1976	S3	Bertunangan	162397000.0	1	1	31	85000	1000	16000	
687	1982	S3	Menikah	160803000.0	0	0	21	55000	16000	1622000	
1300	1971	S2	Bertunangan	157733000.0	1	0	37	39000	1000	9000	
164	1973	S3	Menikah	157243000.0	0	1	98	20000	2000	1582000	
1653	1977	S1	Bertunangan	157146000.0	0	0	13	1000	0	1725000	
2132	1949	S3	Menikah	156924000.0	0	0	85	2000	1000	2000	
655	1975	S1	Cerai	153924000.0	0	0	81	1000	1000	1000	
1898	1945	S3	Lajang	113734000.0	0	0	9	6000	2000	3000	
646	1970	S1	Bertunangan	105471000.0	0	0	36	1009000	181000	104000	

Findings:

- Based on the box plot, the outliers start from the **150 Million income**.
- Thus, there are **eight** outliers.
- From eight people, there are:
 - **five people** have become customer for one year.
 - **One person** had become customer for two years.
 - **Two people** had enrolled under one year.
- The person with income **666 Million** will be considered as **anomaly** and will be **dropped**.

Handling Outliers - Product categories feature

Output:

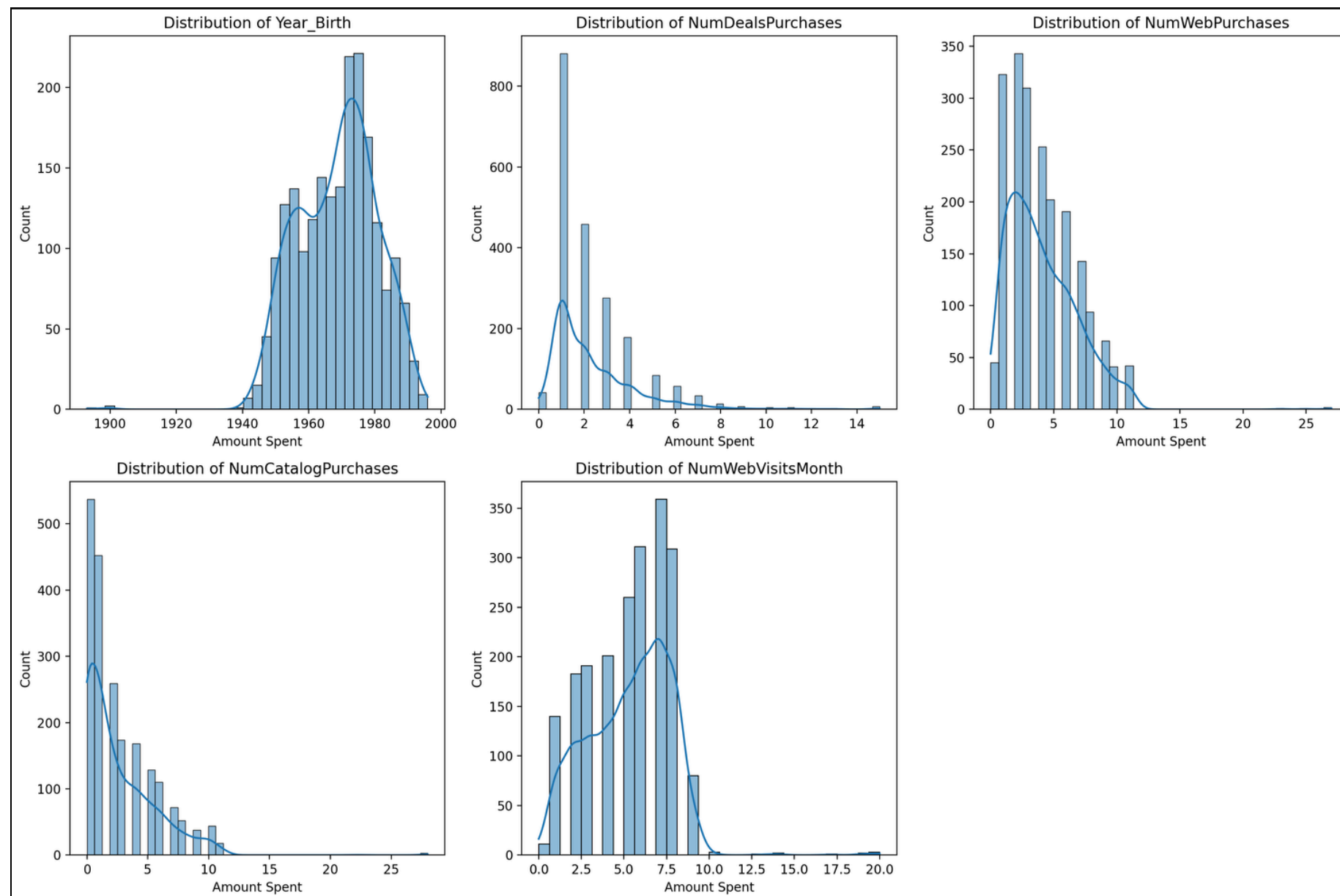


Handling Outliers - Product categories feature

- All product purchase categories exhibit a ***right-skewed*** distribution, indicating that a subset of customers **purchase volumes** significantly **higher** than the **average**.
- The identified outliers **do not represent anomalies**. Therefore, we will apply a **log transformation** to normalize the data distribution.
- This transformation **ensures** that the machine learning model accurately interprets feature importance, **minimizing the risk of bias** caused by skewed data.

Handling Outliers - Product categories feature

Output:



- Most purchase-related features exhibit a **right-skewed** distribution, with many customers making **low to moderate purchases** and a few making significantly higher purchases.
- **Year of Birth** shows a more **symmetric**, approximately normal distribution, so no transformation is applied to this feature.
- To address skewness and improve model performance, **log transformation** was applied to all purchase-related features (NumDealsPurchases, NumWebPurchases, NumCatalogPurchases, NumWebVisitsMonth).
- This transformation helps normalize the data, reducing the impact of extreme values and enabling the machine learning model to better interpret feature importance.

Feature Encoding

Categorical features: Education & Marital_Status

```
['S1' 'S3' 'S2' 'SMA' 'D3']  
['Lajang' 'Bertunangan' 'Menikah' 'Cera' 'Janda' 'Duda']
```

Handling Education feature

- The "Education" feature contains **ordinal** categorical values, meaning the categories have a natural order or ranking (e.g., from high school to doctoral degree).
- To properly represent this order in the model, we suggest using **label encoding**, which assigns integer values to each education level.
- The encoding scheme will start from SMA (Senior High School) as 0, progressing up to S3 (Doctoral degree) as 4, reflecting the increasing level of education.
- This approach preserves the inherent order of education levels, allowing the model to understand the relative ranking rather than treating them as unrelated categories.
- Using label encoding for ordinal features helps improve model accuracy and interpretability compared to one-hot encoding, which treats categories as unordered.

Handling Marital_Status feature

- The "Marital Status" feature contains categorical values such as Lajang (Single), Bertunangan (Engaged), Menikah (Married), Cera' (Divorced), Janda (Widow), and Duda (Widower).
- To **simplify** the categories and **reduce sparsity**, the values Janda (Widow) and Duda (Widower) are **combined** into the Cera' (Divorced) category, as they represent a similar status of no longer being married.
- After this consolidation, **one-hot encoding** is applied to transform the categorical values into binary indicator variables.
- One-hot encoding allows the model to treat each marital status category as a separate feature without implying any ordinal relationship.
- This approach helps the machine learning model effectively use the marital status information without bias or incorrect assumptions about category order.

Feature Engineering

New features:

1. Age

- This feature represents the customer's age, calculated based on their year of birth and the reference year of the dataset.
- Age is a crucial demographic variable that often influences purchasing behavior and preferences.

2. Total_Spending_Log

- This feature is the logarithmically transformed total spending amount, aggregated from multiple purchase categories including Coke, Fruit, Meat, Fish, Sweets, and Gold.
- Applying the log transformation helps normalize the distribution of spending data, reducing the impact of extreme values and making the feature more suitable for modeling.
- This combined spending metric provides a comprehensive view of the customer's overall purchasing power and behavior.

3. Children

- This feature indicates the number of children in the customer's household.
- It helps capture family size and potential influence on purchasing needs and patterns.

4. Is_Parent

- This is a binary feature indicating whether the customer is a parent or not.
- It is derived from the presence of children in the household (e.g., if Children > 0, then Is_Parent = 1; otherwise, 0).
- This feature helps segment customers based on parental status, which can be important for targeted marketing and product recommendations.

Scaling

Standardization on Recency, Total Spending, Enroll tenure, Enroll year, and Age

- Standardization scales features to have a mean of zero and a standard deviation of one, ensuring all features are on the same scale.
- Features like Recency, Total_Spending_log, Enroll_tenure (days), Enroll_year, and Age have different units and value ranges
- Without standardization, features with larger numeric ranges can dominate the distance metric, biasing cluster formation.
- Standardizing these features puts them on the same scale, ensuring each feature contributes equally to the clustering process.

[Link project](#)

[Source Code](#)
[Modelling](#)

Modelling

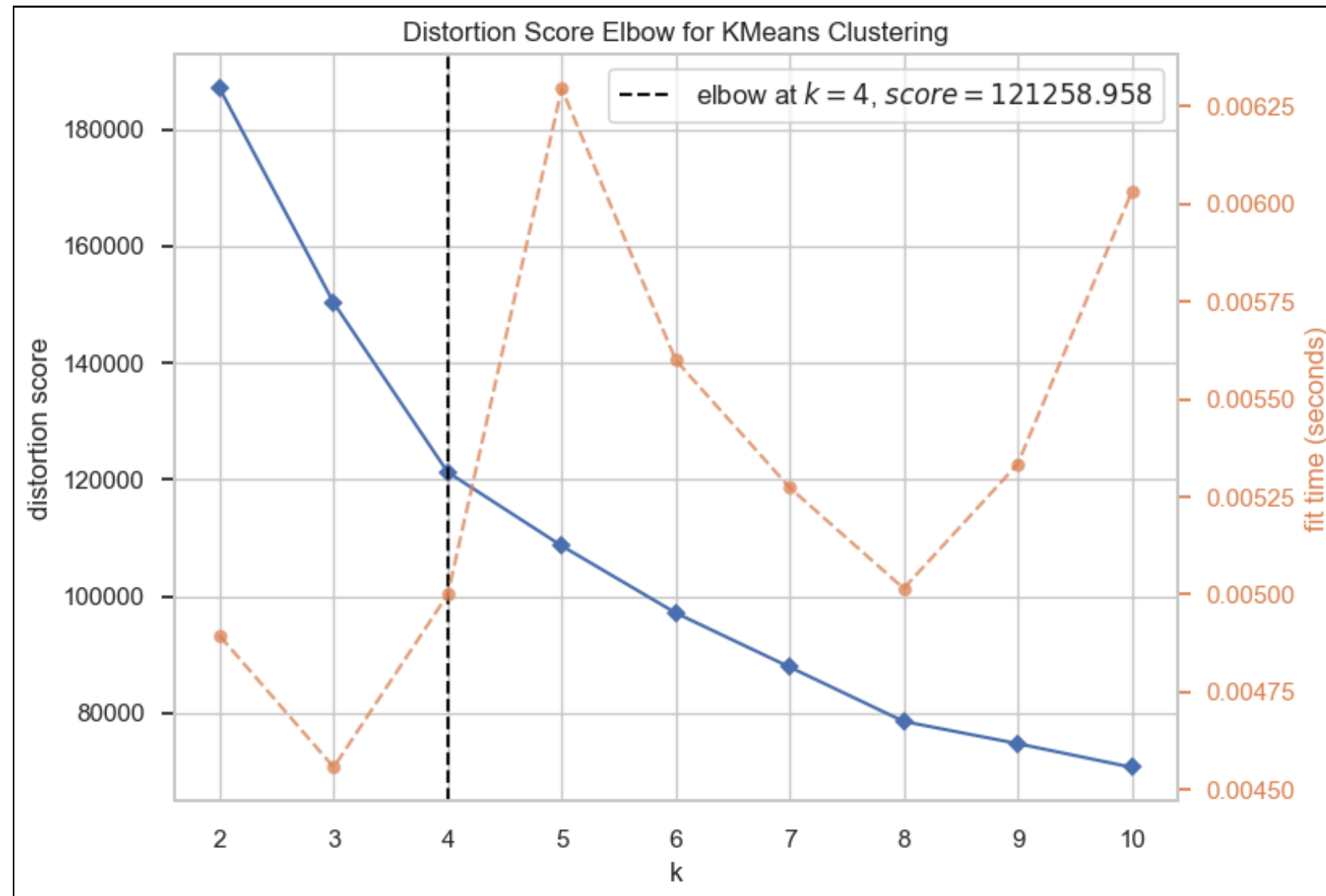
Clustering

Create clusters using Machine Learning

Dataset Information

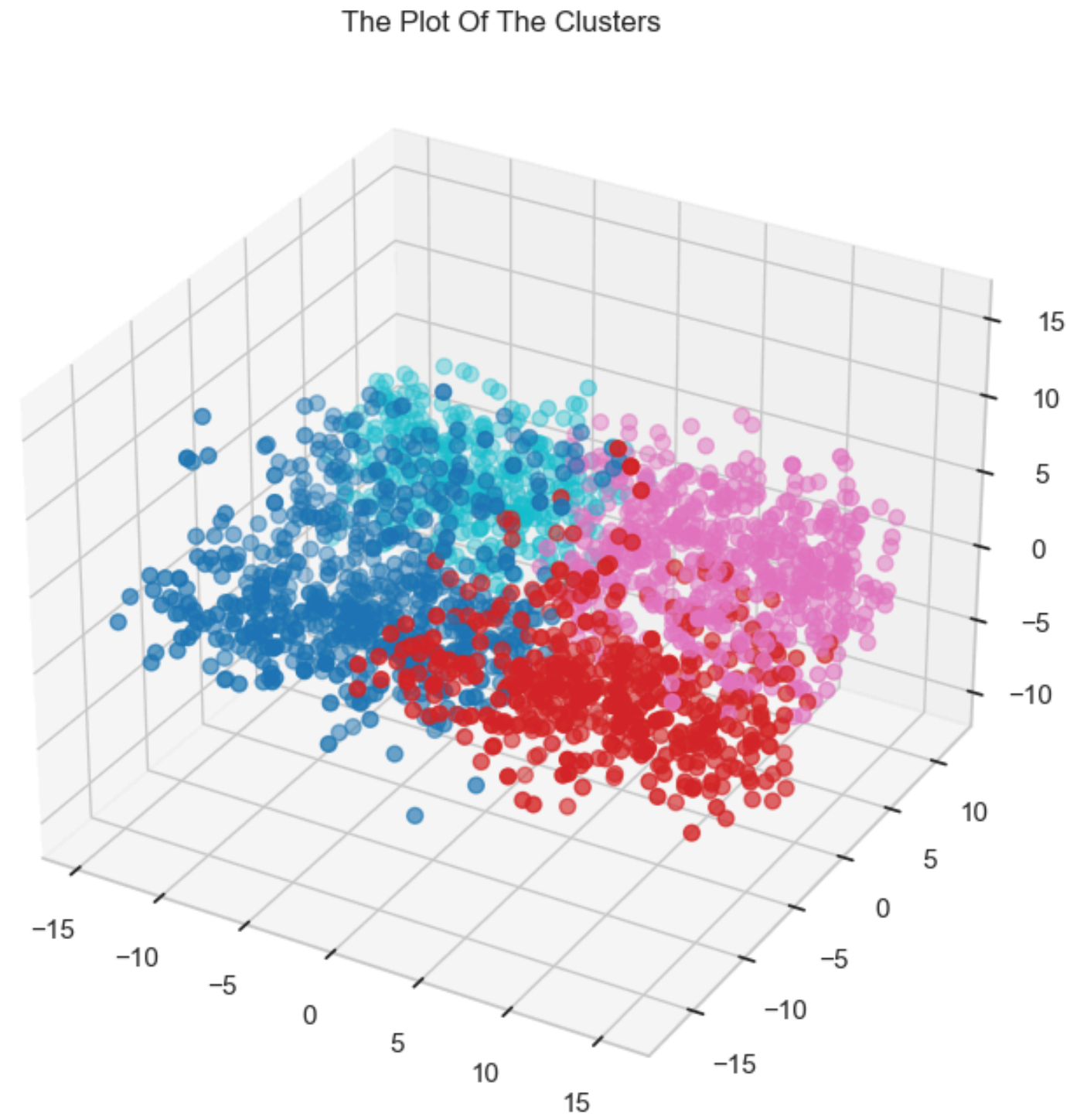
- Dataset contains 2,056 rows and 35 columns (features).
- Many features are correlated, causing redundancy and complexity.
- Dimensionality reduction is needed to simplify the dataset.
- PCA transforms original features into a smaller set of uncorrelated components.
- Unlike feature selection, PCA minimizes information loss while improving interpretability.
- This helps create a simpler, more manageable dataset for analysis and modeling.

Find the best number of clusters

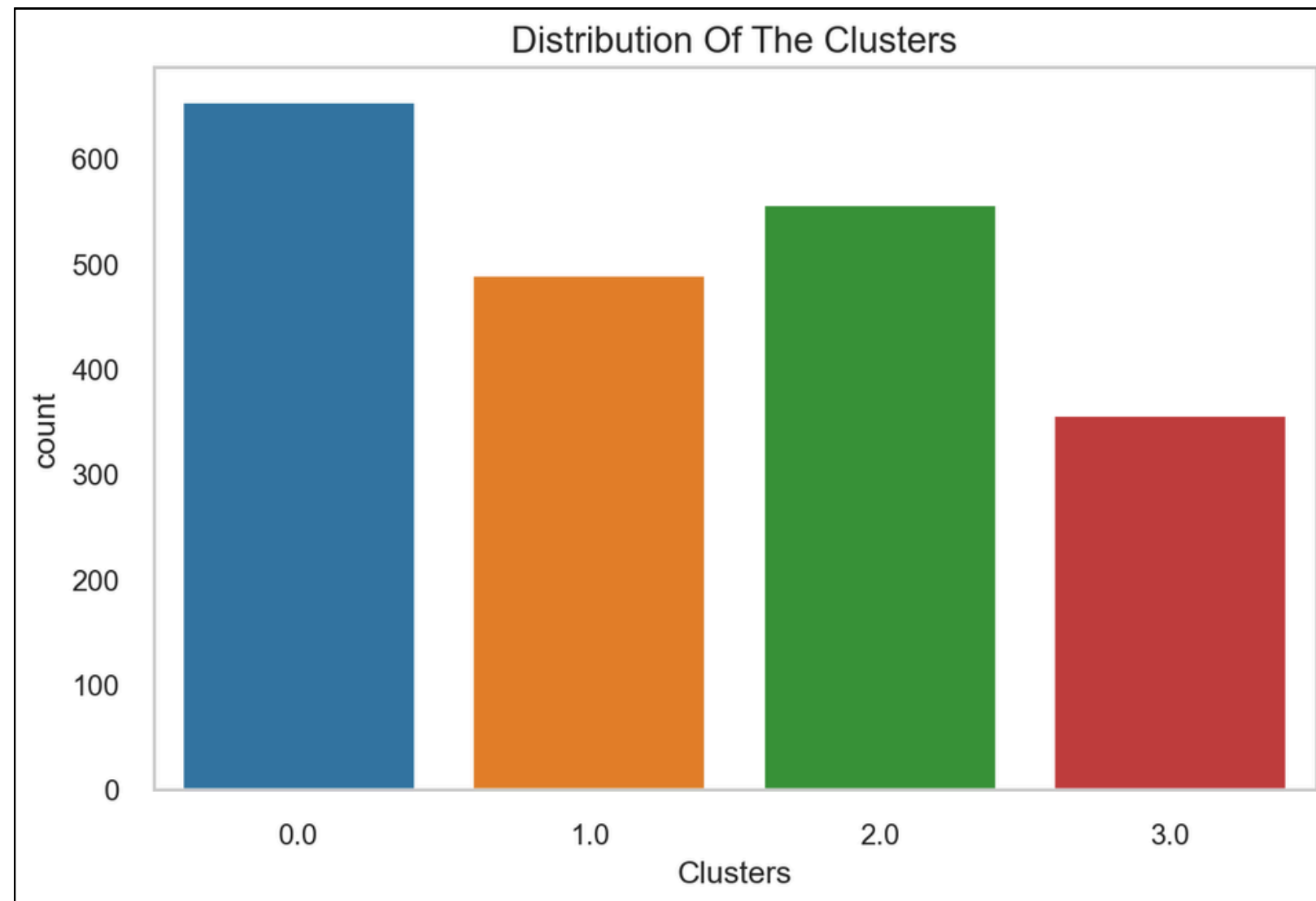


- The elbow plot shows the distortion score for different values of k (number of clusters) in K-Means clustering.
- Distortion score measures the sum of squared distances between data points and their assigned cluster centers; lower values indicate better fit.
- As k increases, distortion decreases, but the rate of improvement slows down.
- The "elbow" point, where the distortion score curve bends, is at $k=4$, indicating an optimal balance between cluster compactness and model simplicity.
- Choosing $k=4$ helps avoid overfitting while capturing meaningful customer segments.
- The plot also shows fit time, which remains relatively stable across different k values, confirming computational efficiency.

The plot of the cluster

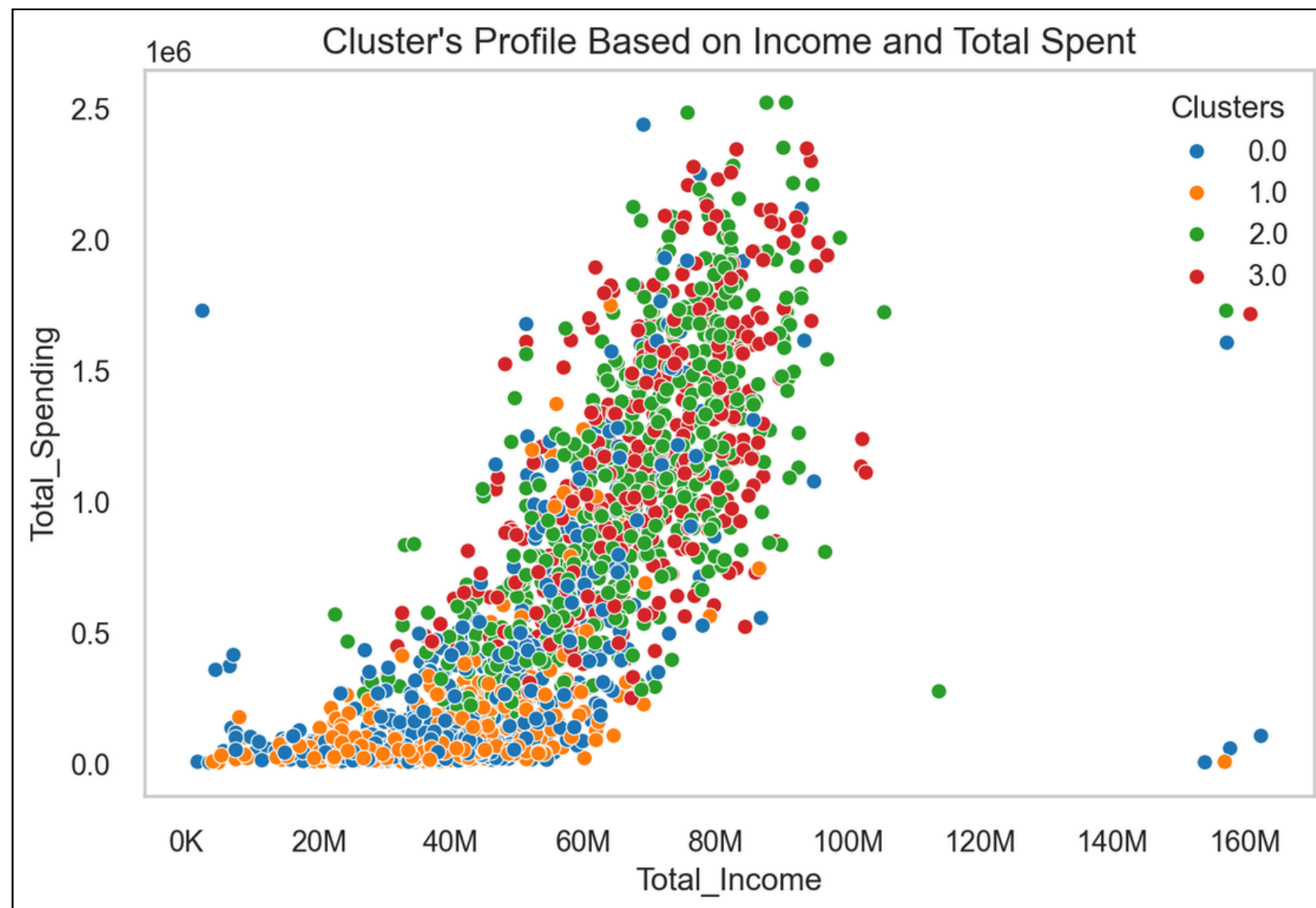


Model Evaluation



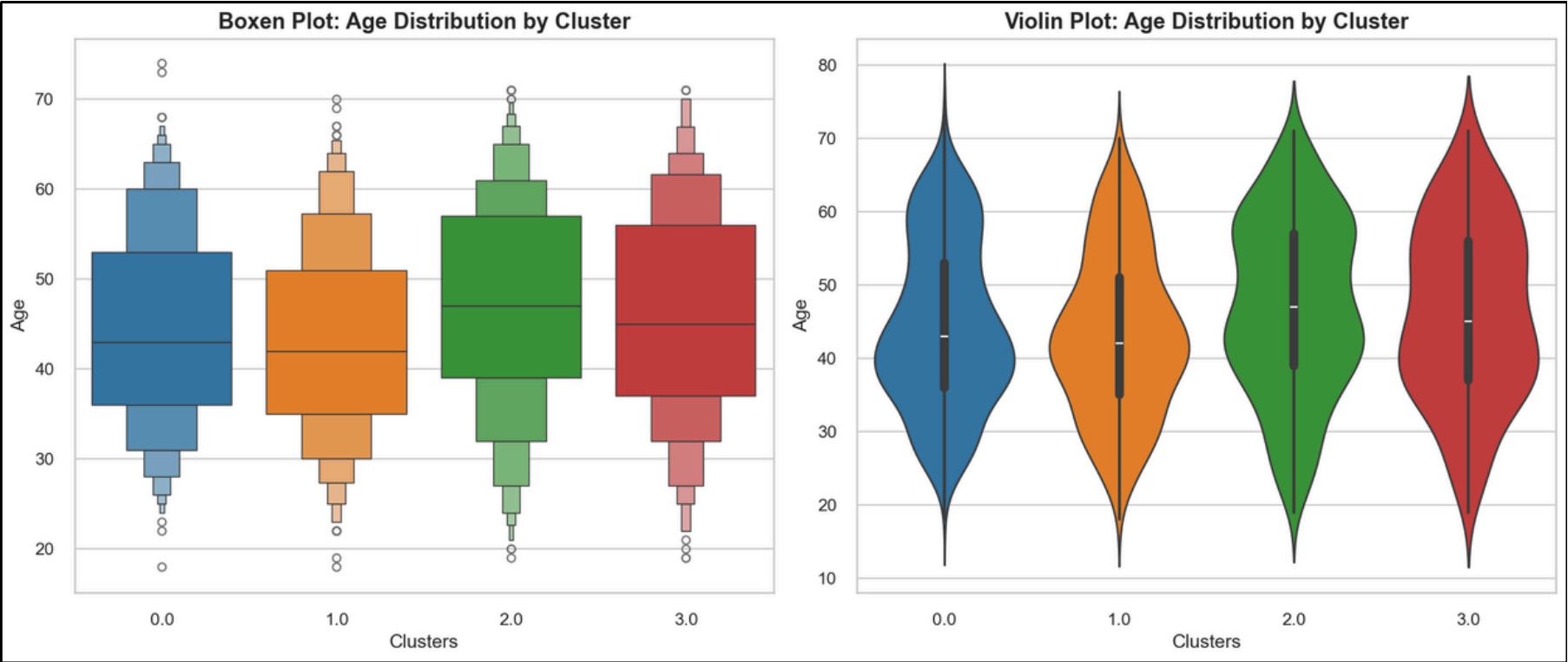
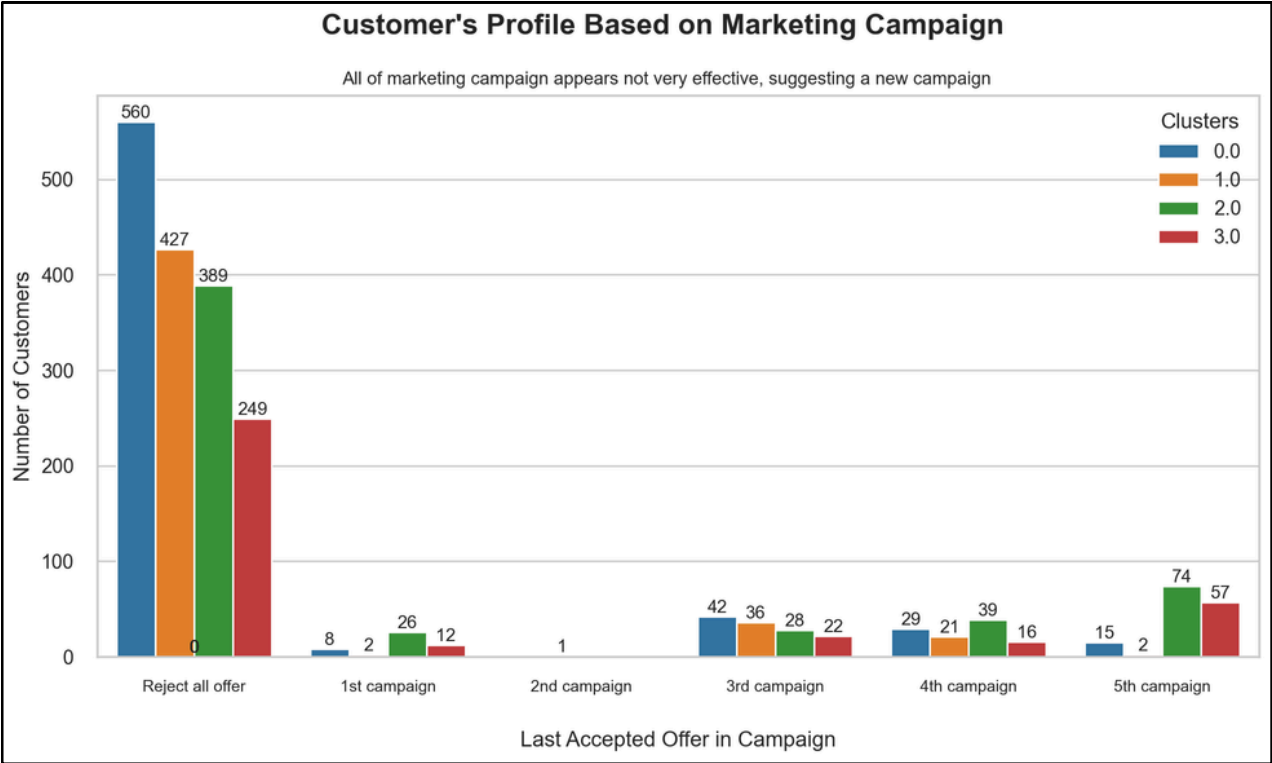
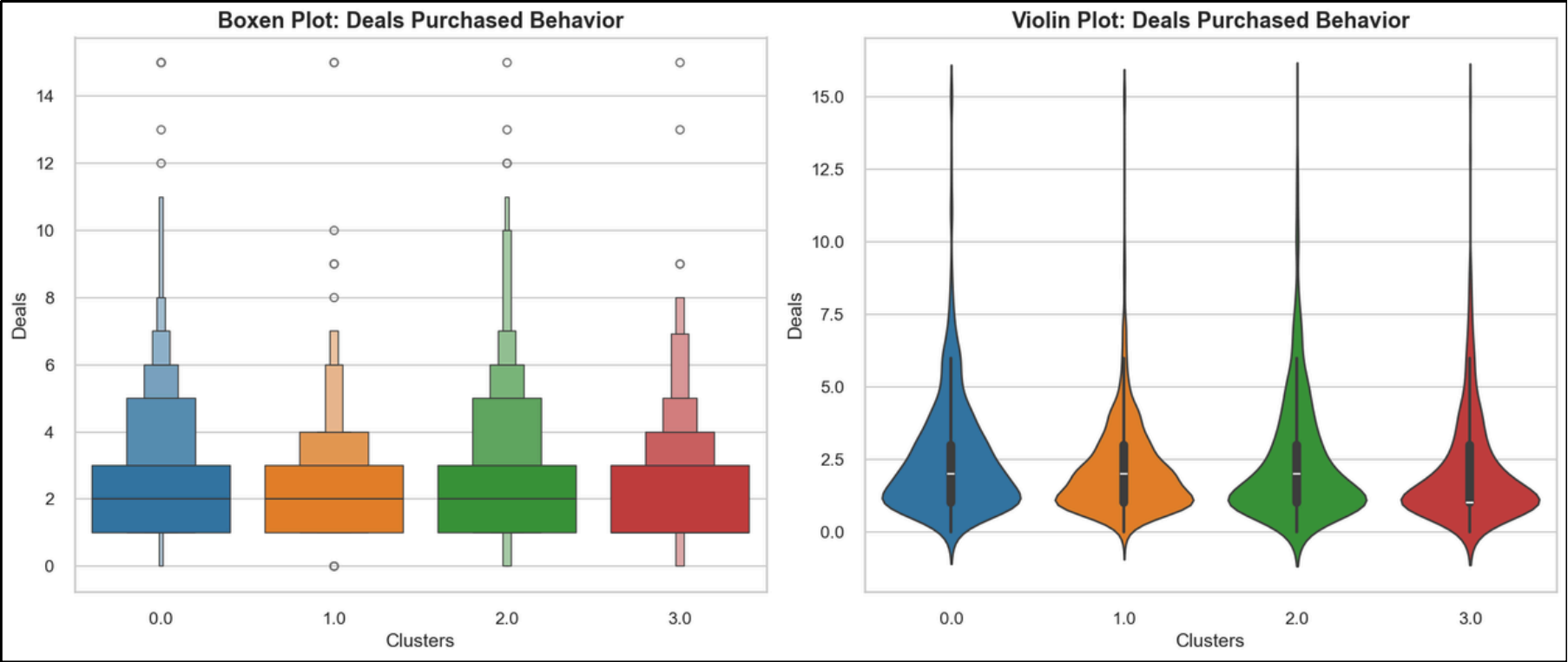
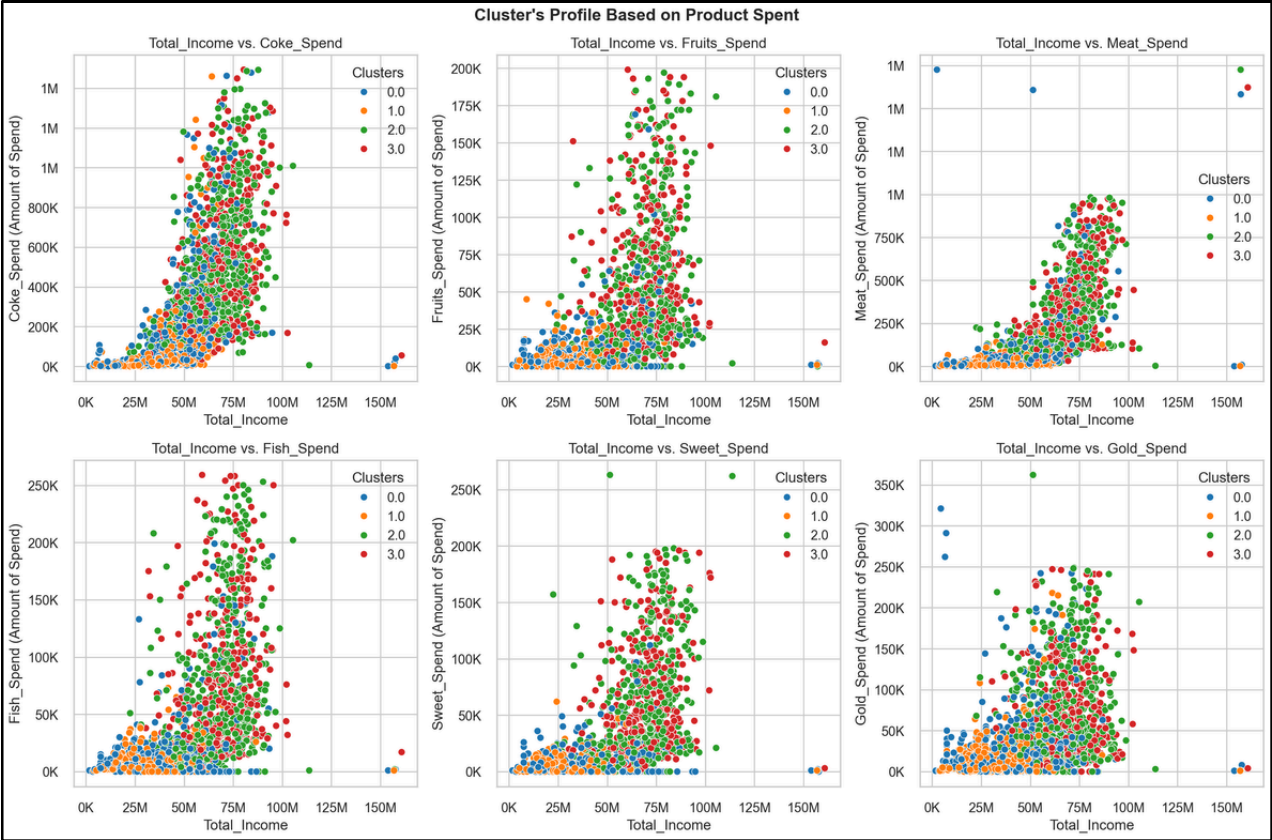
The chart shows the size of each cluster from K-Means. Cluster 0 is the largest, clusters 1 and 2 are medium-sized, and cluster 3 is the smallest. This helps focus marketing on key customer groups.

Clustering based on their income and total purchase

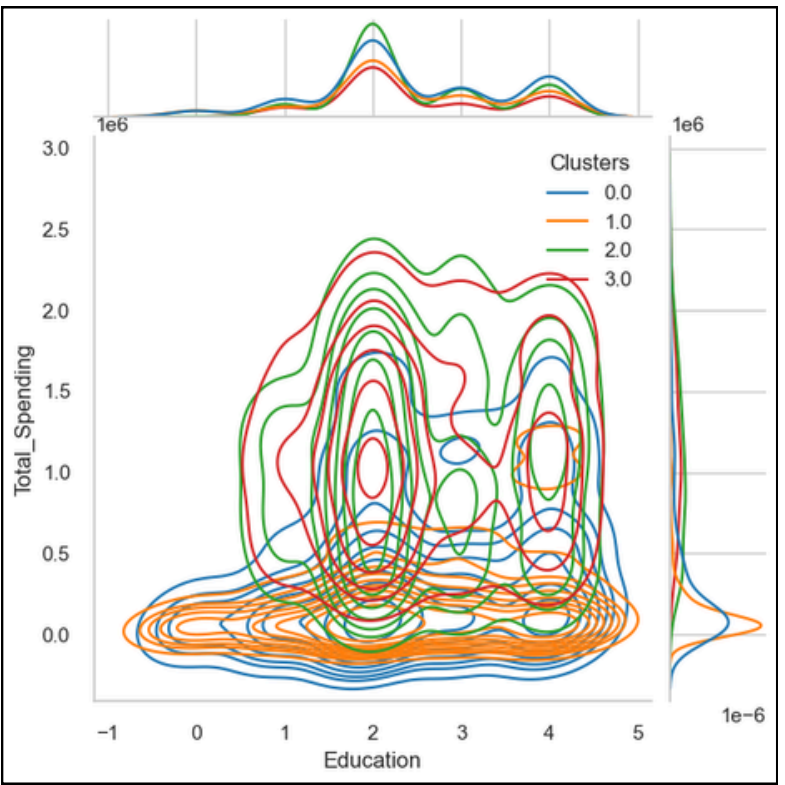
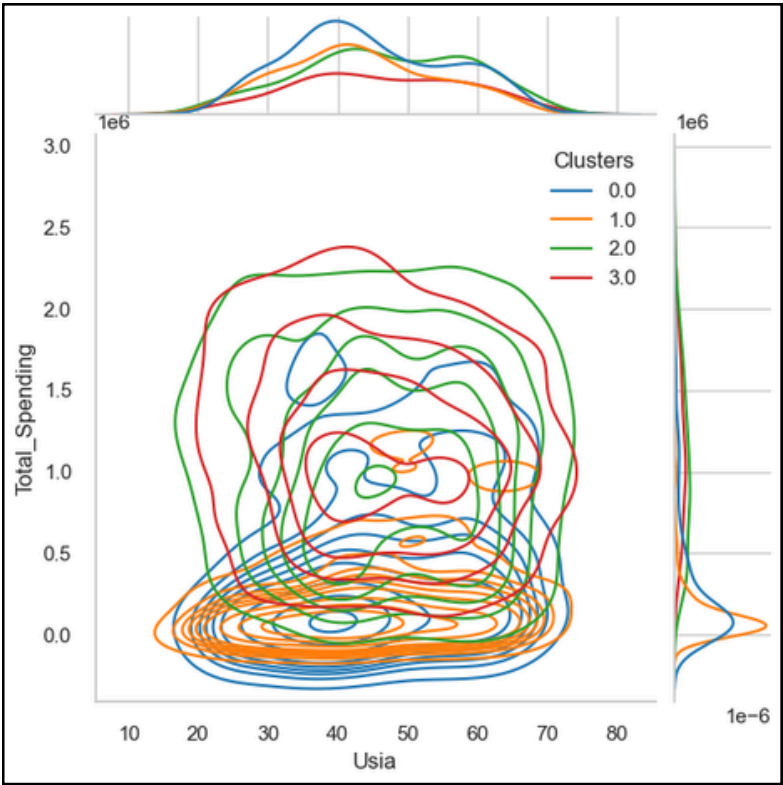
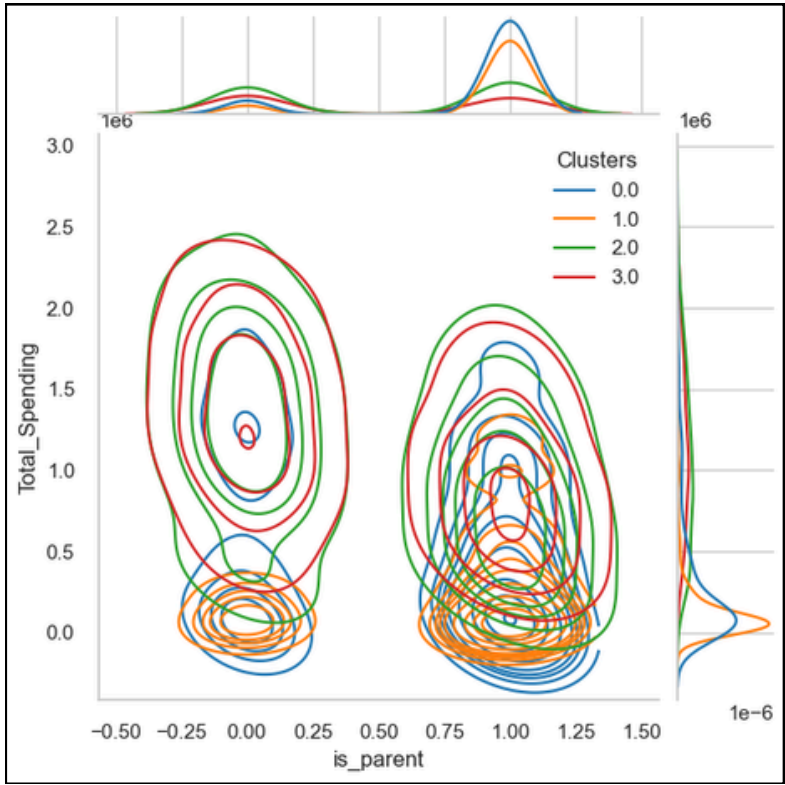
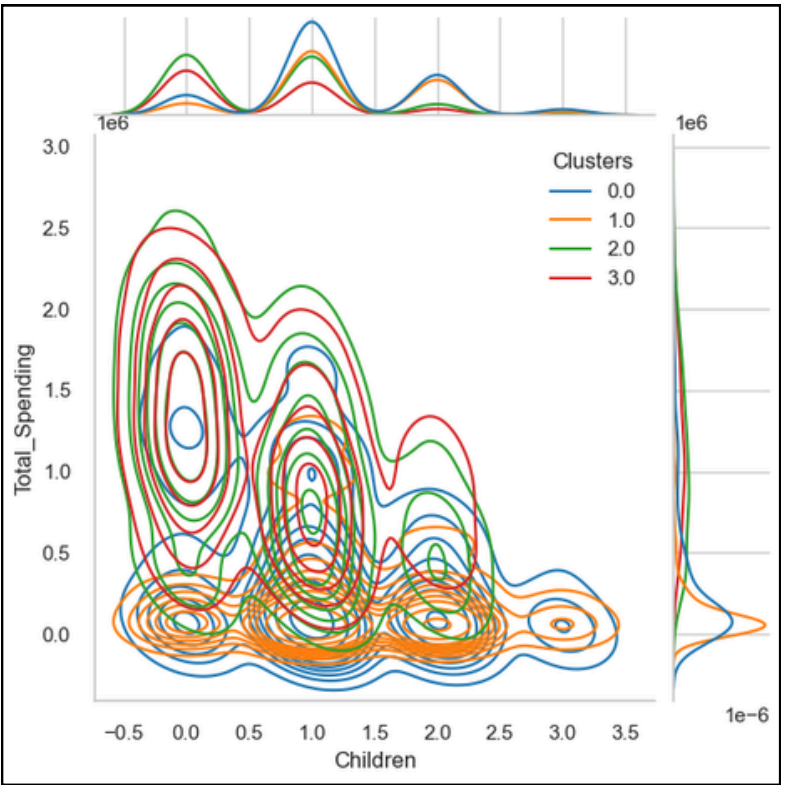
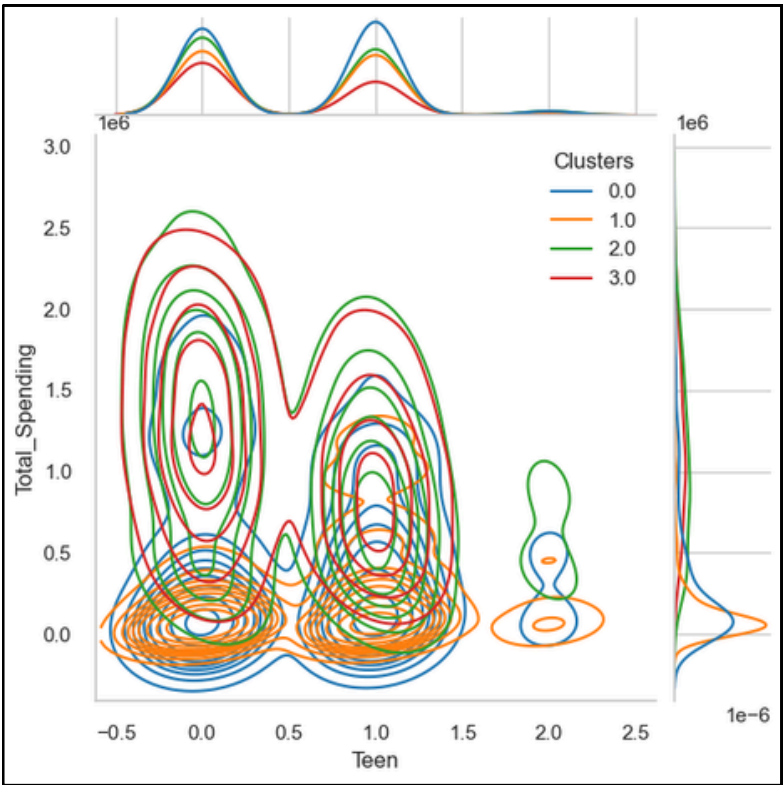
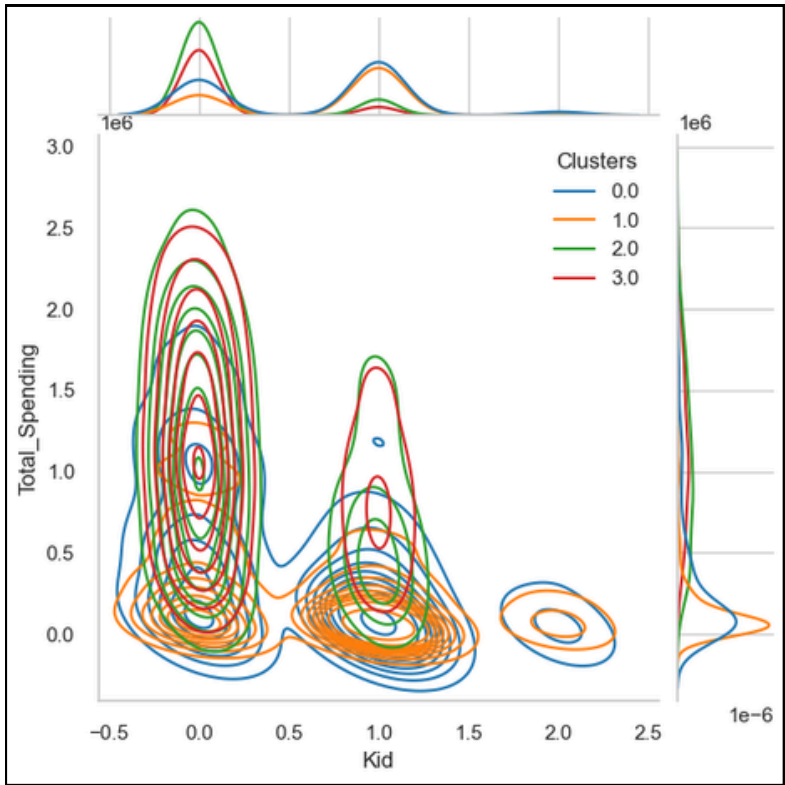


- Cluster 0: Middle-Low Income & Low Spending
- Cluster 1: Low Income & Low Spending
- Cluster 2: Middle-High Income & High Spending
- Cluster 3: High Income & High Spending

Profilling Customers



Profiling Customers



Conclusion

Cluster 0

- Most customers are parents, with some non-parents also showing relatively high spending.
- Typically have one child, with teenagers more common than younger kids.
- Customers aged 30-40 tend to spend the most.

Cluster 1

- Predominantly parents who spend more than non-parents.
- Usually have one child, with teenagers being more prevalent.
- Customers aged 50-60 show higher spending patterns.

Cluster 2

- Parental status is balanced between parents and non-parents, with non-parents tending to spend more.
- Parents mostly have one child, but high spenders also include those without children.
- Teenagers are more common than younger children, with varied age distribution.

Cluster 3

- Mostly non-parents, with the majority having no children.
- Teenagers dominate over younger kids in the household.
- Customers around 40 years old have the highest spending in this group.



The background is a blue-tinted image of a dashboard. It features several data visualizations: a donut chart at the top left with a value of 16.151; a bar chart at the top right with a y-axis labeled 'Count'; a line chart in the middle right with a y-axis labeled 'Total Ice'; and another donut chart at the bottom left with a value of 3,739,958.276. There are also various tables with columns of data visible in the background.

Thanks

[Linkedin Profile](#)

[Github Profile](#)